

URiCA 2026

Upcoming Researchers in
Commutative Algebra

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Conference Notes

URiCA 2026

Continuing the mission of KUMUNUjr

Inspired by the success of the KUMUNU commutative algebra conferences (KU, MU, NU), KUMUNUjr brought together graduate students and postdocs studying commutative algebra in the midwest to share their research and expertise while building lasting collaborations and friendships.

Upcoming Researchers in Commutative Algebra (URiCA) is the next chapter for KUMUNUjr.

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1 New Perspectives on the Briançon–Skoda Theorem

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Abstract

The talk concerned Briançon–Skoda containments from a birational and derived point of view. The running example was $(x^2, y^2) \subseteq \mathbb{C}\{x, y\}$. The main constructions were the normalized blowup, the Buchsbaum–Eisenbud L -complex, birational splitting for pseudo-rational singularities, and a uniform multiplier obtained from alterations.

1.1 Integral closure: the opening example

The first example was $I = (x^2, y^2)$, where integral closure already differs from ordinary ideal membership.

Example 1.1. Consider the convergent power-series ring

$$R = \mathbb{C}\{x, y\}$$

and the ideal

$$I = (x^2, y^2) \subseteq R.$$

Then

$$xy \notin I,$$

while xy belongs to the integral closure of I . Two quick checks were recorded.

(1) **Analytic evidence.** For every $(x, y) \in \mathbb{C}^2$,

$$|xy| \leq \max\{|x|^2, |y|^2\}.$$

Indeed, if $|x| \geq |y|$, then $|xy| \leq |x|^2$, while if $|y| \geq |x|$, then $|xy| \leq |y|^2$.

(2) **Algebraic evidence.** We have

$$(xy)^2 = x^2y^2 \in I^2,$$

since

$$I^2 = (x^2, y^2)(x^2, y^2) = (x^4, x^2y^2, y^4).$$

The appropriate enlargement is the *integral closure* of an ideal.

Definition 1.2 (Integral closure in the analytic local setting). Let

$$J = (f_1, \dots, f_n) \subseteq \mathbb{C}\{x_1, \dots, x_d\}$$

and let $h \in \mathbb{C}\{x_1, \dots, x_d\}$. Then $h \in \bar{J}$ if either—equivalently, both—of the following conditions holds.

(1) **Analytic condition.** There is a constant $C > 0$ such that

$$|h(z)| \leq C \max\{|f_1(z)|, \dots, |f_n(z)|\}$$

for all z in some neighborhood of the origin.

(2) **Algebraic condition.** There exists an integer $r > 0$ and elements

$$a_i \in J^i \quad (1 \leq i \leq r)$$

such that

$$h^r + a_1 h^{r-1} + a_2 h^{r-2} + \dots + a_{r-1} h + a_r = 0. \quad (1.1)$$

Remark 1.3. For $I = (x^2, y^2)$ and $h = xy$, the analytic condition holds with $C = 1$. Algebraically,

$$h^2 - x^2 y^2 = 0,$$

and the constant coefficient $-x^2 y^2$ lies in I^2 . Hence

$$xy \in \bar{I} \setminus I.$$

1.2 Classical Briançon–Skoda containments

The classical Briançon–Skoda containment is the following.

Theorem 1.4 (Briançon–Skoda). *Let*

$$J = (f_1, \dots, f_n) \subseteq \mathbb{C}\{x_1, \dots, x_d\}.$$

Then for every integer $k \geq 1$,

$$\overline{J^{n+k-1}} \subseteq J^k. \quad (1.2)$$

There is also the dimension form

$$\overline{J^{d+k-1}} \subseteq J^k \quad (k \geq 1). \quad (1.3)$$

In particular, taking $k = 1$ in (1.2) gives

$$\overline{J^n} \subseteq J.$$

Remark 1.5. The content is that the integral closure of J^{n+k-1} is still contained in J^k . When both (1.2) and (1.3) apply, one uses the smaller of the two exponent shifts.

The corresponding algebraic statement for regular rings was also recalled.

Theorem 1.6 (Lipman–Sathaye, 1981). *Let R be a regular ring and let*

$$J = (f_1, \dots, f_n) \subseteq R.$$

Then for every $k \geq 1$,

$$\overline{J^{n+k-1}} \subseteq J^k. \quad (1.4)$$

One also has the dimension bound

$$\overline{J^{\dim R + k - 1}} \subseteq J^k \quad (k \geq 1). \quad (1.5)$$

For pseudo-rational singularities the standard formulation uses the dimension.

Theorem 1.7 (Lipman–Teissier, 1981). *Let $(R, \mathfrak{m})$ be a local pseudo-rational ring of dimension d , and let $J \subseteq R$ be any ideal. Then for every $k \geq 1$,*

$$\overline{J^{d+k-1}} \subseteq J^k. \quad (1.6)$$

Remark 1.8. Regular rings admit the sharper bound in terms of the number of generators of J . Pseudo-rational rings may be singular, and the classical Lipman–Teissier statement uses $d = \dim R$. The derived argument below recovers the full n -generated exponent under an appropriate splitting property associated with pseudo-rationality.

1.3 Pseudo-rational singularities and the blowup viewpoint

The examples recorded in the talk were:

Example 1.9 (Examples recorded in the lecture). (1) Regular rings are pseudo-rational.

(2) The quadric cone

$$\mathbb{k}[x, y, z]/(x^2 + y^2 - z^2)$$

is pseudo-rational when $\text{char}(\mathbb{k}) \neq 2$.

(3) The cubic cone

$$\mathbb{k}[x, y, z]/(x^3 + y^3 - z^3)$$

is recorded as never pseudo-rational.

The birational formulation uses a modification on which the ideal is invertible.

1.3.1 Normalized blowup and integral closure

Let $J \subseteq R$ be an ideal, and let

$$\pi: Y \longrightarrow \text{Spec } R$$

be the normalized blowup of J in the setting where this normalization is defined. On the standard affine charts of Y , the pulled-back ideal $J\mathcal{O}_Y$ is principal (equivalently, invertible).

For $I = (x^2, y^2)$ this can be seen directly on the charts.

Example 1.10 (The normalized blowup of (x^2, y^2)). Take

$$I = (x^2, y^2) \subseteq \mathbb{C}[x, y].$$

The normalized blowup is covered by the familiar charts

$$\mathbb{C}\left[\frac{x}{y}, y\right] \quad \text{and} \quad \mathbb{C}\left[x, \frac{y}{x}\right].$$

On the first chart,

$$\begin{aligned} I\mathbb{C}\left[\frac{x}{y}, y\right] &= \left(\left(\frac{x}{y}\right)^2 y^2, y^2\right) \\ &= (y^2), \end{aligned}$$

and

$$xy = \frac{x}{y} y^2 \in (y^2).$$

On the second chart,

$$\begin{aligned} I\mathbb{C}\left[x, \frac{y}{x}\right] &= \left(x^2, \left(\frac{y}{x}\right)^2 x^2\right) \\ &= (x^2), \end{aligned}$$

and

$$xy = \frac{y}{x} x^2 \in (x^2).$$

Thus xy belongs to the pullback of I on every chart of the normalized blowup.

The normalized-blowup criterion was recorded as

$$\bar{J} = \Gamma(Y, J\mathcal{O}_Y) \cap R. \quad (1.7)$$

Remark 1.11 (Normalized blowup). Equation (1.7) is understood inside the total quotient ring (inside the fraction field in the domain case). In Example 1.10, xy is a section of $I\mathcal{O}_Y$ on both charts.

1.4 Derived global sections and the L -complex

For the derived argument one considers

$$\mathbf{R}\Gamma(Y, \mathcal{O}_Y),$$

a complex of R -modules whose cohomology is the sheaf cohomology:

$$H^i(\mathbf{R}\Gamma(Y, \mathcal{O}_Y)) \cong H^i(Y, \mathcal{O}_Y). \quad (1.8)$$

Let

$$J = (f_1, \dots, f_n), \quad \underline{f} = (f_1, \dots, f_n).$$

The quotient R/J is represented by the Koszul complex:

$$H_0(\text{Kos}(\underline{f})) \cong R/J.$$

For higher powers, the analogous replacement is a complex denoted

$$L^k(\underline{f}),$$

with

$$H_0(L^k(\underline{f})) \cong R/J^k. \quad (1.9)$$

Remark 1.12 (Background on $L^k(\underline{f})$). The complex $L^k(\underline{f})$ is the Buchsbaum–Eisenbud L -complex associated to the k th power of the ideal generated by $\underline{f}$. For $k = 1$ it is the Koszul complex. If $f_1, \dots, f_n$ form a regular sequence, $L^k(\underline{f})$ is a free resolution of R/J^k ; in general, the feature used in the lecture is the zeroth-homology identity (1.9).

1.4.1 The derived vanishing on the blowup

Write the pulled-back ideal as

$$J\mathcal{O}_Y = \mathcal{O}_Y(-E),$$

where E denotes the effective exceptional divisor corresponding to J on the blowup.

Lemma 1.13 (The key derived vanishing). *For every $k \geq 1$, the canonical composite*

$$\mathcal{O}_Y(-(n+k-1)E) \longrightarrow \mathcal{O}_Y \longrightarrow L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathcal{O}_Y \quad (1.10)$$

is zero in the derived category $D(Y)$.

Remark 1.14. Here $\mathcal{O}_Y(-(n+k-1)E) = (J\mathcal{O}_Y)^{n+k-1}$. Thus the map to the complex detecting R/J^k vanishes after principalizing J .

Applying derived global sections to (1.10) gives the global form of the same vanishing.

Theorem 1.15 (Global derived Briançon–Skoda mechanism). *The canonical map*

$$\mathbf{R}\Gamma(Y, \mathcal{O}_Y(-(n+k-1)E)) \longrightarrow L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathbf{R}\Gamma(Y, \mathcal{O}_Y) \quad (1.11)$$

is zero in $D(R)$. Consequently, elements of $\overline{J^{n+k-1}}$ map to zero in

$$H_0(L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathbf{R}\Gamma(Y, \mathcal{O}_Y)).$$

Proof. The first assertion is obtained by applying the derived functor $\mathbf{R}\Gamma(Y, -)$ to Lemma 1.13 and using the projection formula. For the second assertion, combine the normalized-blowup description of integral closure with the identification

$$J\mathcal{O}_Y = \mathcal{O}_Y(-E).$$

Thus an element of $\overline{J^{n+k-1}}$ gives a global section of the source of (1.11), hence has zero image in zeroth homology of the target. $\square$

Remark 1.16. The derived vanishing by itself does not yet give membership in J^k . To return to R/J^k one uses a retraction of

$$R \longrightarrow \mathbf{R}\Gamma(Y, \mathcal{O}_Y).$$

1.5 Pseudo-rationality as a splitting mechanism

Pseudo-rationality enters the argument through the following derived splitting condition.

Under the hypotheses used in the talk, pseudo-rationality supplies the birational splitting needed here: for the proper birational models under consideration, the natural map

$$R \longrightarrow \mathbf{R}\Gamma(Y, \mathcal{O}_Y) \quad (1.12)$$

admits a retraction in the derived category. Thus there is a morphism

$$\varphi: \mathbf{R}\Gamma(Y, \mathcal{O}_Y) \longrightarrow R$$

whose composite with (1.12) is the identity. This is the birational derived-splinter property; it is the consequence of pseudo-rationality used in the argument, rather than the definition of pseudo-rationality itself.

The splitting turns the derived vanishing into actual ideal membership.

Theorem 1.17 (Full n -generated Briançon–Skoda from splitting). *Let $J = (f_1, \dots, f_n) \subseteq R$, and let $Y \rightarrow \operatorname{Spec} R$ be the normalized blowup appearing above. Assume that*

$$R \longrightarrow \mathbf{R}\Gamma(Y, \mathcal{O}_Y)$$

admits a retraction in $D(R)$. Then, for every $k \geq 1$,

$$\overline{J^{n+k-1}} \subseteq J^k. \quad (1.13)$$

In particular, the lecture applies this to the pseudo-rational situation.

Proof. Let

$$\varphi: \mathbf{R}\Gamma(Y, \mathcal{O}_Y) \longrightarrow R$$

be a retraction. Tensoring with the perfect complex $L^k(\underline{f})$ gives a map

$$L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathbf{R}\Gamma(Y, \mathcal{O}_Y) \longrightarrow L^k(\underline{f}) \otimes_R^{\mathbf{L}} R.$$

Taking zeroth homology and using (1.9) yields a comparison with R/J^k . Schematically,

$$\begin{array}{ccc} R & \xrightarrow{\quad} & H_0\left(L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathbf{R}\Gamma(Y, \mathcal{O}_Y)\right) \\ & \searrow \text{mod } J^k & \downarrow H_0(1 \otimes \varphi) \\ & & H_0\left(L^k(\underline{f}) \otimes_R^{\mathbf{L}} R\right) \cong R/J^k. \end{array}$$

By Theorem 1.15, every element of $\overline{J^{n+k-1}}$ maps to zero along the top arrow. Therefore its

image in R/J^k is also zero, which is exactly (1.13). $\square$

Remark 1.18. For an n -generated ideal in a d -dimensional pseudo-rational ring, the classical statement in Theorem 1.7 uses the exponent $d + k - 1$. The derived splitting argument recovers the generator-sensitive exponent $n + k - 1$. When $n < d$, this is a genuinely stronger containment.

1.6 Alterations and the uniform Briançon–Skoda direction

The last part of the talk replaced an honest retraction by a retraction up to multiplication. Suppose

$$\eta: R \longrightarrow \mathbf{R}\Gamma(Z, \mathcal{O}_Z)$$

comes from a map

$$Z \longrightarrow \operatorname{Spec} R$$

which may be one of the following objects listed in the lecture:

- a resolution of singularities;
- a regular alteration;
- a truncated regular alteration hypercover.

The construction is required to dominate the blowups above. On derived global sections one has a factorization

$$\begin{array}{ccc} & \mathbf{R}\Gamma(Y, \mathcal{O}_Y) & \\ \nearrow & & \searrow \\ R & \xrightarrow{\eta} & \mathbf{R}\Gamma(Z, \mathcal{O}_Z), \end{array}$$

for the relevant blowups $Y \rightarrow \operatorname{Spec} R$.

Assume in addition that there is a map

$$\phi: \mathbf{R}\Gamma(Z, \mathcal{O}_Z) \longrightarrow R \tag{1.14}$$

such that

$$\phi \circ \eta = c \cdot \operatorname{id}_R \tag{1.15}$$

for some nonzero element $c \in R$.

Remark 1.19. For a pseudo-rational ring one has $c = 1$: the map back is an actual retraction. In the alteration setting one may only be able to split after passing to the generic point. Clearing denominators then produces a nonzero c for which the composite

is multiplication by c . Thus c measures the failure of an honest splitting over all of R .

Combining this “splitting up to c ” with the derived vanishing gives the following containment.

Proposition 1.20 (A uniform multiplier). *Let $J = (f_1, \dots, f_n) \subseteq R$. Under the factorization and map (1.14)–(1.15) described above, the lecture obtains*

$$c \overline{J^{n+k-1}} \subseteq J^k \quad (k \geq 1). \quad (1.16)$$

The important point is that c is chosen from the ambient birational/alteration construction rather than separately from each ideal J .

Proof mechanism. Take $h \in \overline{J^{n+k-1}}$. By Theorem 1.15, its image in

$$H_0(L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathbf{R}\Gamma(Y, \mathcal{O}_Y))$$

is zero. Pass through the factorization to $\mathbf{R}\Gamma(Z, \mathcal{O}_Z)$, then apply $1 \otimes \phi$. Because the composite $R \rightarrow \mathbf{R}\Gamma(Z, \mathcal{O}_Z) \rightarrow R$ is multiplication by c , the image of h in R/J^k becomes the class of ch . Hence $ch = 0$ in R/J^k , or equivalently

$$ch \in J^k.$$

This proves (1.16). □

Remark 1.21. Passing from the existence of a uniform multiplier c in (1.16) to a uniform exponent is not formal; it uses the uniform Briançon–Skoda and Artin–Rees machinery that controls integral closures simultaneously for all ideals.

Corollary 1.22 (Uniform Briançon–Skoda). *Let R be a quasi-excellent reduced Noetherian ring of finite dimension. Then there exists an integer $\ell > 0$, depending only on R , such that for every ideal $J \subseteq R$ and every $k \geq 1$,*

$$\overline{J^{k+\ell}} \subseteq J^k. \quad (1.17)$$

Remark 1.23. The earlier theorems bound the shift by data attached to the particular ideal—the number of generators or the dimension. The uniform statement asserts that one exponent loss ℓ , depending only on the ambient ring, works simultaneously for every ideal. This is the bridge from the Briançon–Skoda problem to uniform Artin–Rees phenomena.

Recap

The running example $xy \in \overline{(x^2, y^2)} \setminus (x^2, y^2)$ was used to pass from the analytic and algebraic definitions of integral closure to the normalized blowup. For $J = (f_1, \dots, f_n)$, the derived vanishing for

$$\mathcal{O}_Y(-(n+k-1)E) \longrightarrow L^k(\underline{f}) \otimes_R^{\mathbf{L}} \mathcal{O}_Y$$

combines with a birational retraction $R \rightarrow \mathbf{R}\Gamma(Y, \mathcal{O}_Y)$ to recover

$$\overline{J^{n+k-1}} \subseteq J^k.$$

Replacing the retraction by a map whose composite is multiplication by a nonzero c gives

$$c \overline{J^{n+k-1}} \subseteq J^k,$$

with c independent of J in the uniform setup. The final uniform Briançon–Skoda statement then uses the corresponding uniform Artin–Rees machinery.

2 The Geometry of the F -Signature

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Joint work in progress with Yuchen Liu

Abstract

The talk reviewed the F -signature and then turned to geometric information carried by F -signature-type invariants for Fano varieties. The material on detecting K -semistability was presented as joint work in progress with Yuchen Liu. The notes record the volume comparison and the examples discussed in the lecture, together with the Artinian Gorenstein viewpoint used in the proof strategy.

2.1 Frobenius and the F -signature

Let k be a perfect field of characteristic $p > 0$, and let $(R, \mathfrak{m}, k)$ be an F -finite local ring essentially of finite type over k . Write

$$d = \dim R.$$

For every integer $e \geq 1$, the e th Frobenius endomorphism is

$$F^e: R \longrightarrow R, \quad r \longmapsto r^{p^e}.$$

The notation $F_*^e R$ denotes the additive group of R endowed with the R -module structure

$$r \cdot F_*^e x = F_*^e(r^{p^e} x).$$

Thus the iterates of Frobenius produce a sequence of finite R -modules

$$R, F_* R, F_*^2 R, \dots$$

Theorem 2.1 (Kunz). *The ring R is regular if and only if $F_*^e R$ is free for some, equivalently for every, $e \geq 1$.*

For a singular ring one measures the free summand of $F_*^e R$.

Definition 2.2 (Frobenius splitting numbers and F -signature). For $e \geq 1$, write

$$F_*^e R \cong R^{\oplus a_e(R)} \oplus M_e,$$

where M_e has no free direct summand and $a_e(R)$ is maximal. The integer $a_e(R)$ is the e th *Frobenius splitting number*. Since the residue field is perfect, the F -signature is

$$s(R) = \lim_{e \rightarrow \infty} \frac{a_e(R)}{p^{ed}}.$$

The existence of this limit is due to Tucker.

Theorem 2.3 (Basic properties of the F -signature). *For an F -finite local ring as above,*

$$0 \leq s(R) \leq 1.$$

Moreover:

- (i) $s(R) = 1$ if and only if R is regular;
- (ii) $s(R) > 0$ if and only if R is strongly F -regular.

Consequently, positivity of the F -signature forces R to be normal and Cohen–Macaulay; in the standard excellent setting it also implies pseudo-rationality.

Example 2.4 (Veronese subrings). Let

$$S = k[x_1, \dots, x_n]$$

and let $A = S^{(r)} \subseteq S$ be the r th Veronese subring. If $\mathfrak{m} = A_+$ is its homogeneous maximal ideal, then in the tame case $p \nmid r$ the corresponding quotient-singularity computation gives

$$s(A_{\mathfrak{m}}) = \frac{1}{r}.$$

Hence the F -signature can be positive while becoming arbitrarily small as $r \rightarrow \infty$.

Example 2.5 (The Fermat cubic cone). For a prime p , set

$$A_p = \frac{\mathbb{F}_p[x, y, z, w]}{(x^3 + y^3 + z^3 + w^3)}, \quad \mathfrak{m} = (x, y, z, w)A_p, \quad R_p = (A_p)_{\mathfrak{m}}.$$

For $p = 2, 3$ one has $s(R_p) = 0$. For $p \geq 5$, write $p = 3k + 1$ or $p = 3k + 2$. Then

$$s(R_p) = \frac{1}{8} - \frac{1}{4(27k^2 + 27k + 8)}.$$

In particular,

$$\lim_{p \rightarrow \infty} s(R_p) = \frac{1}{8}.$$

The projective variety

$$X_p = \text{Proj}(A_p)$$

is a smooth cubic surface for $p \neq 3$. This example already indicates that the asymptotic behavior of the local invariant $s(R_p)$ can reflect global projective geometry.

2.2 Fano varieties and anticanonical section rings

Definition 2.6 (Fano variety). Let X be a smooth projective variety of dimension n over a field. It is *Fano* if the anticanonical line bundle

$$\omega_X^{-1} \cong \bigwedge^n T_X$$

is ample.

Example 2.7 (Hypersurfaces). Let

$$X = V(f) \subseteq \mathbb{P}^{n+1}$$

be a smooth hypersurface defined by a homogeneous polynomial of degree d . Adjunction gives

$$K_X \cong \mathcal{O}_X(d - n - 2).$$

Thus:

- X is Fano when $d \leq n + 1$;
- $K_X \cong \mathcal{O}_X$ when $d = n + 2$;
- K_X is ample when $d \geq n + 3$.

For a Fano variety X , consider its anticanonical section ring

$$A(X, -K_X) = \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(-mK_X)).$$

When discussing its F -signature, we localize at the homogeneous maximal ideal

$$\mathfrak{m} = A(X, -K_X)_+$$

and write

$$R = A(X, -K_X)_{\mathfrak{m}}.$$

We also set

$$\mathrm{vol}(X) = (-K_X)^n.$$

Theorem 2.8 (Volume bound recorded in the talk). *Under the hypotheses on the Fano variety and anticanonical section ring used in the lecture, let X have dimension n , let $A = A(X, -K_X)$, and put $R = A_{A_+}$. The stated inequality is*

$$\mathrm{vol}(X) \geq 2^n(n+1)! s(R).$$

Equivalently,

$$s(R) \leq \frac{\mathrm{vol}(X)}{2^n(n+1)!}.$$

The equality case is the geometric focus.

Theorem 2.9 (Liu–Pande, work in progress). *In the special situation considered in the talk, let X be an n -dimensional Fano variety in positive characteristic, let $A = A(X, -K_X)$, and put $R = A_{A_+}$. If*

$$s(R) = \frac{\mathrm{vol}(X)}{2^n(n+1)!},$$

then X is K -semistable.

The conference abstract describes the K -semistability detection as holding in special situations; the formulation above records the version presented in these notes and should be read as work in progress rather than as an unrestricted published theorem.

2.3 The F -normalized volume

The lecture also used a Frobenius analogue of normalized volume.

Definition 2.10 (F -normalized volume). Let $(R, \mathfrak{m})$ be a d -dimensional local ring of characteristic $p > 0$. Define

$$\widehat{\mathrm{vol}}_F(R) = \inf_{\substack{\mathfrak{a} \subseteq \mathfrak{m} \\ \mathfrak{a} \text{ is } \mathfrak{m}\text{-primary}}} \mathrm{fpt}(\mathfrak{a})^d e(\mathfrak{a}),$$

where $\mathrm{fpt}(\mathfrak{a})$ is the F -pure threshold and $e(\mathfrak{a})$ is the Hilbert–Samuel multiplicity.

The F -pure threshold measures how far powers of an ideal can be inserted into Frobenius

splittings. It plays the positive-characteristic role of the log canonical threshold. The corresponding quantity

$$\mathrm{fpt}(\mathfrak{a})^d e(\mathfrak{a})$$

is the Frobenius analogue of the expressions appearing in the normalized-volume theory of singularities. The comparison with K -stability is therefore parallel to the characteristic-zero criterion that detects K -semistability from the singularity of the anticanonical cone.

2.4 Applications

Example 2.11 (Full flag varieties). Let V be an n -dimensional vector space over k , and let

$$\mathrm{Fl}(V)$$

be the full flag variety parametrizing chains

$$0 = V_0 \subset V_1 \subset \cdots \subset V_n = V, \quad \dim V_i = i.$$

The full flag variety is Fano. For its anticanonical section ring R , the F -signature attains the extremal value

$$s(R) = \frac{\mathrm{vol}(\mathrm{Fl}(V))}{2^{\dim \mathrm{Fl}(V)} (\dim \mathrm{Fl}(V) + 1)!}.$$

The equality criterion therefore recovers the K -semistability of the flag variety.

Example 2.12 (The Fermat cubic surface). Let

$$X = V(x^3 + y^3 + z^3 + w^3) \subseteq \mathbb{P}^3.$$

Then X is a del Pezzo surface of degree 3, so

$$\mathrm{vol}(X) = (-K_X)^2 = 3.$$

Hence

$$\frac{\mathrm{vol}(X)}{2^2 \cdot 3!} = \frac{3}{24} = \frac{1}{8}.$$

Together with

$$\lim_{p \rightarrow \infty} s(R_p) = \frac{1}{8},$$

this shows that the reductions of the Fermat cubic asymptotically attain the geometric upper bound for the F -signature.

2.5 Proof ideas: Artinian Gorenstein algebras

A key mechanism is to convert free summands of Frobenius into finite-length Gorenstein quotients.

Definition 2.13 (Artinian Gorenstein algebra). A finite-dimensional local k -algebra $(A, \mathfrak{n})$ is Artinian Gorenstein if

$$\dim_k \operatorname{Ann}_A(\mathfrak{n}) = 1.$$

Equivalently, its socle is one-dimensional over k .

In the standard graded case,

$$A = A_0 \oplus A_1 \oplus \cdots \oplus A_j, \quad A_0 \cong A_j \cong k,$$

Artinian Gorenstein duality gives

$$\dim_k A_i = \dim_k A_{j-i}$$

for every i . Thus the Hilbert function is symmetric.

For a local Artinian Gorenstein algebra, the associated graded ring

$$\operatorname{gr}_{\mathfrak{n}}(A) = A/\mathfrak{n} \oplus \mathfrak{n}/\mathfrak{n}^2 \oplus \mathfrak{n}^2/\mathfrak{n}^3 \oplus \cdots$$

need not itself be Gorenstein. Nevertheless, self-duality gives, for every ideal $I \subseteq A$,

$$\ell(A/I) = \ell(\operatorname{Ann}_A(I)).$$

If $\mathfrak{n}^{j+1} = 0$, then

$$\mathfrak{n}^{j+1-i} \subseteq \operatorname{Ann}_A(\mathfrak{n}^i),$$

which yields strong “front-loaded” constraints on the Hilbert function.

The Frobenius splitting numbers admit precisely such an Artinian interpretation.

Definition 2.14 (Frobenius splitting ideal). For $e \geq 1$, define

$$I_e(R) = \{x \in R \mid \phi(F_*^e x) \in \mathfrak{m} \text{ for every } \phi \in \operatorname{Hom}_R(F_*^e R, R)\}.$$

Proposition 2.15 (Gorenstein reduction). Assume that $(R, \mathfrak{m}, k)$ is F -finite and Goren-

stein, with k perfect. Then $R/I_e(R)$ is Artinian Gorenstein and

$$\ell_R(R/I_e(R)) = a_e(R).$$

Thus estimates for the Frobenius splitting numbers can be converted into length and Hilbert-function estimates for Artinian Gorenstein quotients.

These finite-length Gorenstein quotients are the local input for the asymptotic F -signature estimates used in the volume comparison.

3 Systems of Higher Homotopies and Koszul Homomorphisms

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Abstract

The talk recast systems of higher homotopies as twisting morphisms into an endomorphism DG algebra. For complete intersections this recovers the Eisenbud–Shamash construction; the same language extends to Koszul homomorphisms through the Priddy coalgebra. Exact zero-divisors were used as the main non-complete-intersection example.

3.1 Systems of higher homotopies as twisting morphisms

Let F be a complex of R -modules and let C be a differential graded coalgebra. The endomorphism complex

$$\mathrm{End}_R(F)$$

is a differential graded algebra under composition.

Definition 3.1 (System of higher homotopies). A *system of higher homotopies* on F over C is a twisting morphism

$$\sigma: C \longrightarrow \mathrm{End}_R(F).$$

In the curved setting, the coalgebra C is equipped with a degree -2 R -linear map

$$h: C \longrightarrow R,$$

called the *curvature*. If

$$\eta_{\mathrm{End}}: R \longrightarrow \mathrm{End}_R(F), \quad r \longmapsto r \, \mathrm{id}_F,$$

denotes the unit map, then σ has degree -1 and satisfies the curved Maurer–Cartan equation

$$d_{\mathrm{End}}\sigma + \sigma d_C + \mu_{\mathrm{End}}(\sigma \otimes \sigma)\Delta_C = \eta_{\mathrm{End}} \circ h,$$

with the standard Koszul signs.

For complete intersections this is the usual Shamash–Eisenbud higher-homotopy data.

3.2 The classical complete-intersection construction

Let

$$S = R/(f_1, \dots, f_c),$$

where $f_1, \dots, f_c$ is an R -regular sequence, and let

$$F \xrightarrow{\sim} M$$

be a free R -resolution of an S -module M . Write

$$H = (h_1, \dots, h_c) \in \mathbb{N}^c, \quad |H| = h_1 + \dots + h_c,$$

and let e_i denote the i th standard basis vector of $\mathbb{N}^c$.

Definition 3.2 (Classical higher homotopies). A classical system of higher homotopies on F consists of maps

$$\sigma^{(H)}: F \longrightarrow F$$

of homological degree $2|H| - 1$, indexed by $H \in \mathbb{N}^c$, satisfying

$$\begin{aligned} \sigma^{(0)} &= \partial^F, \\ \partial^F \sigma^{(e_i)} + \sigma^{(e_i)} \partial^F &= f_i \operatorname{id}_F, \\ \partial^F \sigma^{(H)} + \sigma^{(H)} \partial^F &= - \sum_{\substack{I+J=H \\ I, J \neq 0}} \sigma^{(I)} \sigma^{(J)} \quad (|H| \geq 2). \end{aligned}$$

Thus $\sigma^{(e_i)}$ is a null-homotopy for multiplication by f_i ; the higher maps supply the remaining coherence relations.

Example 3.3. Let

$$R = k[a, b, c], \quad S = R/(a^2, b^2), \quad M = k,$$

and take the Koszul resolution

$$F = \operatorname{Kos}^R(a, b, c).$$

If e_1, e_2, e_3 are the exterior generators, define

$$\sigma^{(1,0)} = a e_1 \wedge (-), \quad \sigma^{(0,1)} = b e_2 \wedge (-).$$

Then

$$\partial^F \sigma^{(1,0)} + \sigma^{(1,0)} \partial^F = a^2 \operatorname{id}_F,$$

and similarly for b^2 . Since $e_1^2 = e_2^2 = 0$ and $e_1 e_2 = -e_2 e_1$, all higher maps $\sigma^{(H)}$ with $|H| \geq 2$ may be taken to be zero.

3.3 Resolution transfer

Let

$$\Gamma = \Gamma_R(y_1, \dots, y_c)$$

be the divided-power coalgebra on generators y_i of degree 2. Its basis consists of

$$y^{(H)} = y_1^{(h_1)} \cdots y_c^{(h_c)}, \quad H \in \mathbb{N}^c,$$

with comultiplication

$$\Delta(y^{(H)}) = \sum_{I+J=H} y^{(I)} \otimes y^{(J)}.$$

The dual polynomial variables act on Γ by contraction.

Theorem 3.4 (Shamash–Eisenbud resolution transfer). *Let $S = R/(f_1, \dots, f_c)$ be a quotient by an R -regular sequence, and let $F \rightarrow M$ be an R -free resolution of an S -module M equipped with a system of higher homotopies $\{\sigma^{(H)}\}$. Then the graded S -module*

$$S \otimes_R \Gamma \otimes_R F$$

admits a differential obtained by combining the contraction operators on Γ with the maps $\sigma^{(H)}$, and the resulting complex is an S -free resolution of M .

Concretely, if t^H denotes the contraction operator dual to $y^{(H)}$, the differential has the schematic form

$$d = \sum_{H \in \mathbb{N}^c} t^H \otimes \sigma^{(H)}.$$

The higher-homotopy identities are exactly what is needed for $d^2 = 0$.

3.4 Koszul duality and the Priddy coalgebra

Definition 3.5 (Koszul DG algebra). Let A be an augmented DG k -algebra. It is *Koszul* if the residue field k admits a linear semifree resolution over A .

Polynomial rings, exterior algebras, and quadratic monomial algebras provide basic examples. For a Koszul algebra A , the linear resolution of k can be encoded by a coalgebra C , called the *Priddy coalgebra*, together with a twisting morphism

$$\tau: C \longrightarrow A.$$

In the DG setting relevant here, C also carries a degree -2 curvature map

$$h: C \longrightarrow R.$$

If

$$\eta: R \longrightarrow A$$

is the unit map, the twisting morphism satisfies the corresponding curved Maurer–Cartan equation, whose right-hand side is $\eta \circ h$. The corresponding twisted tensor product

$$A \otimes_{\tau} C$$

is a resolution of k .

Example 3.6 (Polynomial rings). Let

$$A = k[a_1, \dots, a_n].$$

Then

$$C = \bigwedge_k (e_1, \dots, e_n)$$

with

$$\Delta(e_i) = 1 \otimes e_i + e_i \otimes 1$$

and the usual shuffle comultiplication on higher wedges. Define

$$\tau(e_i) = a_i,$$

and let τ vanish on wedge degree different from 1. The twisted tensor product

$$A \otimes_{\tau} C$$

is precisely the Koszul complex

$$\mathrm{Kos}^A(a_1, \dots, a_n).$$

Example 3.7 (The Koszul complex of a sequence). Let

$$A = \mathrm{Kos}^R(f_1, \dots, f_c).$$

The relevant coalgebra is the divided-power coalgebra

$$C = \Gamma_R(y_1, \dots, y_c),$$

with

$$\Delta(y^{(H)}) = \sum_{I+J=H} y^{(I)} \otimes y^{(J)}.$$

If e_i is the i th Koszul generator, the twisting morphism is determined in lowest degree by

$$\tau(y_i) = e_i.$$

The curvature is determined on the degree-2 cogenerators by

$$h(y_i) = d_A(e_i) = f_i,$$

and is zero on the remaining divided-power basis elements. Under this identification, a classical system of higher homotopies becomes a twisting morphism

$$\sigma: C \longrightarrow \text{End}_R(F).$$

Evaluating the curved Maurer–Cartan equation on the primitive element y_i gives

$$\partial^F \sigma(y_i) + \sigma(y_i) \partial^F = f_i \text{id}_F,$$

which is precisely the second condition in the classical Eisenbud–Shamash definition. Thus the curvature term records the differentials of the degree-1 generators of A , while the remaining components of the Maurer–Cartan equation recover the higher coherence relations.

3.5 Koszul homomorphisms and universal resolution transfer

The same construction makes sense beyond complete intersections.

Theorem 3.8 (Resolution transfer for Koszul homomorphisms). *Let*

$$\varphi: R \longrightarrow S$$

be a Koszul homomorphism, let M be an S -module, and let C be the Priddy coalgebra associated with φ . If

$$F \xrightarrow{\sim} M$$

is a free R -resolution equipped with a system of higher homotopies

$$\sigma: C \longrightarrow \text{End}_R(F),$$

then the twisted complex built on

$$S \otimes_R C \otimes_R F$$

is an S -free resolution of M .

For a quotient by a regular sequence this is the classical Shamash–Eisenbud construction; for general Koszul homomorphisms the Priddy coalgebra gives the corresponding transfer complex.

3.6 Exact zero-divisors

Definition 3.9 (Exact pair of zero-divisors). Let R be a commutative Noetherian ring. Elements $a, b \in R$ form an *exact pair of zero-divisors* if

$$\text{Ann}_R(a) = (b), \quad \text{Ann}_R(b) = (a).$$

Let $S = R/(a)$. For an exact pair (a, b) , the homomorphism

$$R \longrightarrow S$$

is Koszul. The associated Priddy coalgebra has the form

$$C = \Gamma_R(y) \otimes_R \bigwedge_R(x),$$

where

$$|y| = 2, \quad |x| = 3,$$

and

$$\begin{aligned} \Delta(y^{(n)}) &= \sum_{i=0}^n y^{(i)} \otimes y^{(n-i)}, \\ \Delta(x) &= 1 \otimes x + x \otimes 1. \end{aligned}$$

The natural basis is given by the elements

$$y^{(n)} \quad \text{and} \quad xy^{(n)}.$$

Here the degree-2 cogenerator carries the curvature

$$h(y) = a,$$

corresponding to the differential of the degree-1 generator in the Koszul-type DG algebra

associated with $R \rightarrow S$; the curvature vanishes on the other cogenerators.

Let F be an R -free resolution of an S -module. A system of higher homotopies can then be written as maps

$$\alpha^n, \beta^n: F \longrightarrow F,$$

where

$$|\alpha^n| = 2n - 1, \quad |\beta^n| = 2n + 2.$$

The twisting-morphism equation becomes the following recursive system:

$$\begin{aligned} \alpha^0 &= \partial^F, \\ \partial^F \alpha^1 + \alpha^1 \partial^F &= a \operatorname{id}_F, \\ \partial^F \alpha^n + \alpha^n \partial^F &= - \sum_{i=1}^{n-1} \alpha^i \alpha^{n-i} \quad (n \geq 2), \\ \partial^F \beta^n - \beta^n \partial^F &= (n+1)b \alpha^{n+1} - \sum_{i=1}^{n-1} (\beta^i \alpha^{n-i} + \alpha^i \beta^{n-i}). \end{aligned}$$

The last family is the extra coherence relation forced by the degree-3 exterior generator. In this way the coalgebraic definition recovers the explicit higher homotopies attached to exact zero-divisors.

Recap

A system of higher homotopies is encoded by a twisting morphism

$$C \longrightarrow \operatorname{End}_R(F).$$

For complete intersections, C is the divided-power coalgebra and the resulting twisted complex is the Shamash–Eisenbud resolution. For a Koszul homomorphism, the Priddy coalgebra gives the same resolution-transfer mechanism. In the exact-zero-divisor example, the degree-2 divided-power generator and degree-3 exterior generator produce the explicit α^n - and β^n -recursions above.

4 Edge Ideals via Independence Polynomials of Graphs

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Joint work with Jennifer Biermann, Trung Chau, Selvi Kara, Augustine O'Keefe, Joseph Skelton, Gabriel Sosa Castillo, and with Takayuki Hibi and Selvi Kara

Abstract

The talk concerned the relation between edge ideals and independence polynomials of graphs. The main result expresses the degree of the h -polynomial of $S/I(G)$ in terms of the multiplicity of -1 as a root of the independence polynomial $P_G(x)$. The discussion also included consequences for regularity and pseudo-Gorenstein properties, together with graph operations such as suspension.

4.1 Independence polynomials and edge ideals

The basic dictionary is

$$\text{graph } G \longleftrightarrow \text{edge ideal } I(G).$$

The talk used the independence polynomial $P_G(x)$ to read off information about the Hilbert series of $S/I(G)$. The main invariant is

$$M(G) = \text{mult}_{x=-1} P_G(x),$$

and the degree formula proved in the joint work is

$$\deg h_{S/I(G)}(t) = \alpha(G) - M(G).$$

4.2 Edge Ideals and Algebraic Invariants

4.2.1 Edge ideals

Let

$$G = (V, E), \quad V = \{x_1, \dots, x_n\},$$

be a finite simple graph and let

$$S = k[x_1, \dots, x_n]$$

be a polynomial ring over a field k .

Definition 4.1 (Edge ideal). The *edge ideal* of G is the squarefree quadratic monomial ideal

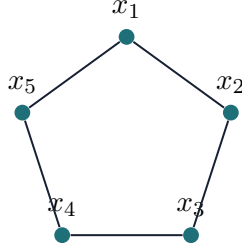
$$I(G) = (x_i x_j : \{x_i, x_j\} \in E(G)) \subseteq S.$$

Example 4.2 (The 5-cycle). For the cycle

$$C_5 : \quad x_1 - x_2 - x_3 - x_4 - x_5 - x_1,$$

one has

$$I(C_5) = (x_1x_2, x_2x_3, x_3x_4, x_4x_5, x_5x_1) \subseteq k[x_1, x_2, x_3, x_4, x_5].$$



4.2.2 Minimal graded free resolutions

Let $I \subseteq S$ be a homogeneous ideal. A minimal graded free resolution of S/I has the form

$$0 \longrightarrow \bigoplus_{j \in \mathbb{Z}} S(-j)^{\beta_{p,j}} \longrightarrow \cdots \longrightarrow \bigoplus_{j \in \mathbb{Z}} S(-j)^{\beta_{1,j}} \longrightarrow S \longrightarrow S/I \longrightarrow 0.$$

The graded Betti number $\beta_{i,j}(S/I)$ is the number of copies of $S(-j)$ appearing in homological degree i .

These numbers determine, among other invariants,

$$\text{pdim}(S/I) = \max\{i : \beta_{i,j} \neq 0 \text{ for some } j\},$$

and

$$\text{reg}(S/I) = \max\{j - i : \beta_{i,j} \neq 0\}.$$

Example 4.3 (The path P_3). Consider

$$I(P_3) = (x_1x_2, x_2x_3) \subseteq k[x_1, x_2, x_3].$$

For this displayed ideal, the minimal resolution is

$$0 \longrightarrow S(-3) \xrightarrow{\begin{bmatrix} -x_3 \\ x_1 \end{bmatrix}} S(-2)^2 \xrightarrow{\begin{bmatrix} x_1x_2 & x_2x_3 \end{bmatrix}} S \longrightarrow S/I(P_3) \longrightarrow 0.$$

Thus

$$\beta_{1,2} = 2, \quad \beta_{2,3} = 1.$$

4.3 Independence Complexes and Hochster's Formula

4.3.1 The independence complex

Definition 4.4 (Independence complex). The *independence complex* of G is

$$\Delta(G) = \{F \subseteq V(G) : F \text{ is an independent set of } G\}.$$

The independence number $\alpha(G)$ is the maximum cardinality of an independent set of G . Consequently,

$$\dim \Delta(G) = \alpha(G) - 1.$$

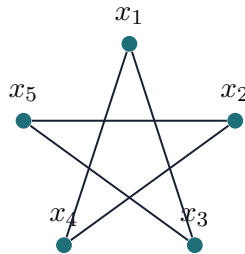
For $G = C_5$, the independent sets are

$$\emptyset, \quad \{x_1\}, \dots, \{x_5\}, \quad \{x_1, x_3\}, \{x_1, x_4\}, \{x_2, x_4\}, \{x_2, x_5\}, \{x_3, x_5\}.$$

Thus

$$\alpha(C_5) = 2,$$

and the 1-skeleton of $\Delta(C_5)$ is again a 5-cycle; with the chosen placement of the vertices, it appears as a pentagram.



4.3.2 Stanley–Reisner translation

The Stanley–Reisner ideal of $\Delta(G)$ is generated by the minimal nonfaces of $\Delta(G)$.

Since the minimal nonfaces of an independence complex are precisely the edges of G ,

$$I_{\Delta(G)} = I(G).$$

This permits topological information about induced subcomplexes of $\Delta(G)$ to control the Betti numbers of the edge ideal.

Theorem 4.5 (Hochster's formula). For $W \subseteq V(G)$, let $\Delta(G)_W$ denote the induced subcomplex on W .

Then

$$\beta_{i,j}(S/I_{\Delta(G)}) = \sum_{\substack{W \subseteq V(G) \\ |W|=j}} \dim_k \widetilde{H}_{j-i-1}(\Delta(G)_W; k).$$

Because

$$I_{\Delta(G)} = I(G),$$

this computes the graded Betti numbers of $S/I(G)$ combinatorially.

4.3.3 Fröberg's theorem as a model application

A central example of the combinatorics–algebra correspondence is the following result quoted in the talk.

Theorem 4.6 (Fröberg, 1990). *For a graph G with at least one edge,*

$$\operatorname{reg}(S/I(G)) = 1 \iff \overline{G} \text{ is chordal.}$$

Here chordal means that every cycle of length at least 4 has a chord; equivalently, the graph has no induced cycle of length at least 4.

For example, consider the graph whose edge ideal is

$$I(G) = (x_1x_3, x_3x_5, x_1x_4) \subseteq k[x_1, x_2, x_3, x_4, x_5].$$

Its complement is chordal, hence

$$\operatorname{reg}(S/I(G)) = 1.$$

The broader point is that expressing graded Betti numbers in terms of $\Delta(G)$ gives a powerful method for reading algebraic information from graph-theoretic or topological data.

4.4 Hilbert Series and the h -Polynomial

4.4.1 Hilbert series from the resolution

The Hilbert series of $S/I(G)$ is

$$H_{S/I(G)}(t) = \sum_{d \geq 0} \dim_k (S/I(G))_d t^d.$$

The graded free resolution yields the expression

$$H_{S/I(G)}(t) = \frac{\sum_{i \geq 0} \left(\sum_{j \in \mathbb{Z}} (-1)^i \beta_{i,j} t^j \right)}{(1-t)^n}.$$

This expression need not yet be reduced.

Since

$$\dim(S/I(G)) = \alpha(G),$$

we may write the Hilbert series in reduced form as

$$H_{S/I(G)}(t) = \frac{h_{S/I(G)}(t)}{(1-t)^{\alpha(G)}},$$

where $h_{S/I(G)}(t)$ is the h -polynomial.

For brevity, write

$$h_G(t) = h_{S/I(G)}(t) = \sum_{i \geq 0} h_i(G) t^i.$$

4.4.2 The f -vector and h -vector

If

$$f(\Delta(G)) = (f_{-1}, f_0, \dots, f_{\alpha(G)-1}),$$

then f_{i-1} counts independent sets of cardinality i .

In particular, for C_5 ,

$$f(\Delta(C_5)) = (1, 5, 5).$$

The corresponding h -polynomial is

$$h_{S/I(C_5)}(t) = 1 + 3t + t^2,$$

so its h -vector is

$$(1, 3, 1).$$

Determining the entire h -vector can be difficult. A coarser but still meaningful invariant is

$$\deg h_{S/I(G)}(t) = \max\{i : h_i \neq 0\}.$$

4.4.3 What was known about $\deg h$

If S/I is Cohen–Macaulay, then

$$\operatorname{reg}(S/I) = \deg h_{S/I}(t).$$

Without the Cohen–Macaulay hypothesis, the two invariants may behave independently. The following prior results motivate the problem.

- **Hibi–Matsuda (2018).** For every pair of integers $r, d \geq 1$, there is a monomial ideal I for which

$$\operatorname{reg}(S/I) = r, \quad \deg h_{S/I}(t) = d.$$

- **Hibi–Matsuda–Van Tuyl (2019).** For every pair $r, d \geq 1$, such a pair is also realized by an edge ideal. Moreover, if G has n vertices, then

$$\operatorname{reg}(S/I(G)) + \deg h_{S/I(G)}(t) \leq n.$$

- **Biermann–Kara–O’Keefe–Skelton–Sosa Castillo (2025).** Explicit formulas for $\deg h$ were obtained for several graph families, including paths, cycles, and complete bipartite graphs.

The goal is to express

$$\deg h_{S/I(G)}(t)$$

entirely in terms of combinatorial data from an arbitrary graph G .

4.5 From Independent Sets to the Degree of the h -Polynomial

4.5.1 Expressing the h -polynomial using $\Delta(G)$

Let g_i denote the number of independent sets of G of size i .

Then

$$g_i = f_{i-1} \quad (0 \leq i \leq \alpha(G)),$$

with

$$g_0 = 1$$

corresponding to the empty set.

The f -vector-to- h -vector relation gives

$$h_{S/I(G)}(t) = \sum_{i=0}^{\alpha(G)} g_i t^i (1-t)^{\alpha(G)-i}. \quad (4.1)$$

In particular,

$$\deg h_{S/I(G)}(t) \leq \alpha(G).$$

4.5.2 The top coefficient and the reduced Euler characteristic

The coefficient in top possible degree satisfies

$$h_{\alpha(G)} = (-1)^{\alpha(G)-1} \tilde{\chi}(\Delta(G)),$$

where

$$\tilde{\chi}(\Delta(G)) = \sum_{i=0}^{\alpha(G)} (-1)^{i-1} f_{i-1} = \sum_{i=0}^{\alpha(G)} (-1)^{i-1} g_i.$$

Thus maximal degree is controlled by an alternating sum of the numbers of independent sets.

4.5.3 The independence polynomial

Definition 4.7 (Independence polynomial). The *independence polynomial* of G is

$$P_G(x) = \sum_{i=0}^{\alpha(G)} g_i x^i.$$

For example,

$$P_{C_5}(x) = 1 + 5x + 5x^2.$$

The previous alternating sum is precisely the special value at -1 :

$$\tilde{\chi}(\Delta(G)) = -P_G(-1).$$

Consequently,

$$h_{\alpha(G)} = (-1)^{\alpha(G)} P_G(-1),$$

and hence

$$\deg h_{S/I(G)}(t) = \alpha(G) \iff P_G(-1) \neq 0. \quad (4.2)$$

4.5.4 Multiplicity of the root -1

When

$$P_G(-1) = 0,$$

write

$$P_G(x) = (1+x)^{M(G)} Q(x), \quad Q(-1) \neq 0,$$

where

Definition 4.8.

$$M(G) = \text{ord}_{x=-1} P_G(x)$$

is the multiplicity of $x = -1$ as a root of the independence polynomial.

The main theorem identifies the exact amount by which the degree drops from its maximal possible value.

Theorem 4.9 (Biermann–Chau–Kara–O’Keefe–Skelton–Sosa Castillo–Vien). *For every finite simple graph G ,*

$$\deg h_{S/I(G)}(t) = \alpha(G) - M(G).$$

Thus each additional order of vanishing of $P_G(x)$ at $x = -1$ lowers the degree of the h -polynomial by one.

Since

$$\dim(S/I(G)) = \alpha(G),$$

the theorem also yields

$$a(S/I(G)) = \deg h_{S/I(G)}(t) - \dim(S/I(G)) = -M(G).$$

The same change-of-variables argument applies more generally to Stanley–Reisner rings: the degree drop is governed by the order of vanishing at -1 of the face-enumerating polynomial. In particular, the graph statement has a squarefree-monomial, or hypergraph, analogue.

4.5.5 Proof of the degree formula

Proof. Starting from (4.1), factor the expression through the independence polynomial:

$$\begin{aligned} h_{S/I(G)}(t) &= \sum_{i=0}^{\alpha(G)} g_i t^i (1-t)^{\alpha(G)-i} \\ &= (1-t)^{\alpha(G)} P_G\left(\frac{t}{1-t}\right). \end{aligned}$$

Set

$$u = \frac{1}{1-t}, \quad \frac{t}{1-t} = u - 1.$$

Then

$$h_{S/I(G)}(t) = u^{-\alpha(G)} P_G(u - 1).$$

If

$$P_G(x) = (1+x)^{M(G)} Q(x), \quad Q(-1) \neq 0,$$

then

$$h_{S/I(G)}(t) = u^{M(G)-\alpha(G)} Q(u - 1).$$

Equivalently, with

$$d = \alpha(G) - M(G),$$

we obtain

$$h_{S/I(G)}(t) = (1-t)^d Q\left(\frac{t}{1-t}\right).$$

Because

$$\deg P_G = \alpha(G),$$

one has

$$\deg Q = d.$$

Writing

$$Q(x) = \sum_{i=0}^d q_i x^i,$$

gives

$$h_{S/I(G)}(t) = \sum_{i=0}^d q_i t^i (1-t)^{d-i}.$$

The coefficient of t^d is

$$\sum_{i=0}^d q_i (-1)^{d-i} = (-1)^d Q(-1),$$

which is nonzero.

Hence

$$\deg h_{S/I(G)}(t) = d = \alpha(G) - M(G).$$

□

4.5.6 Examples

Example 4.10 (The cycle C_5). We have

$$P_{C_5}(x) = 1 + 5x + 5x^2.$$

Since

$$P_{C_5}(-1) = 1,$$

one has

$$\alpha(C_5) = 2, \quad M(C_5) = 0.$$

Therefore

$$\deg h_{S/I(C_5)}(t) = 2.$$

Example 4.11 (The path P_4). A direct count gives

$$P_{P_4}(x) = 1 + 4x + 3x^2 = (1 + x)(1 + 3x).$$

Thus

$$\alpha(P_4) = 2, \quad M(P_4) = 1,$$

and Theorem 4.9 gives

$$\deg h_{S/I(P_4)}(t) = 1.$$

4.6 Pseudo-Gorenstein and Pseudo-Gorenstein*

4.6.1 Related notions

Let A be a standard graded Cohen–Macaulay k -algebra with h -vector

$$(h_0, h_1, \dots, h_s).$$

If A is Gorenstein, then its h -vector is symmetric and in particular $h_s = 1$. The pseudo-Gorenstein condition isolates this last numerical feature: in the Cohen–Macaulay setting, A is called *pseudo-Gorenstein* when its last nonzero h -coefficient is 1.

For edge ideals, Hibi–Kara–Vien consider the corresponding graph condition and single out the extremal case in which the h -polynomial has the largest possible degree.

Definition 4.12 (Pseudo-Gorenstein* graph; Hibi–Kara–Vien). A graph G is called *pseudo-Gorenstein** if

$$\deg h_G(t) = \alpha(G) \quad \text{and} \quad h_{\alpha(G)}(G) = 1.$$

Equivalently, the a -invariant of $S/I(G)$ is 0 and the top h -coefficient is 1.

Using

$$h_{\alpha(G)} = (-1)^{\alpha(G)} P_G(-1),$$

we obtain the following criterion.

Proposition 4.13 (Hibi–Kara–Vien criterion). *A graph G is pseudo-Gorenstein* if and only if*

$$P_G(-1) = (-1)^{\alpha(G)}.$$

4.6.2 Paths and cycles

The independence polynomial satisfies the standard vertex recursion

$$P_G(x) = P_{G-v}(x) + xP_{G-N[v]}(x), \quad (4.3)$$

where $N[v]$ is the closed neighborhood of v .

In particular,

$$P_{P_n}(x) = P_{P_{n-1}}(x) + xP_{P_{n-2}}(x),$$

and

$$P_{C_n}(x) = P_{P_{n-1}}(x) + xP_{P_{n-3}}(x).$$

At $x = -1$, the path values become periodic. The beginning of the sequence is

n	1	2	3	4	5	6	7
$P_{P_n}(-1)$	0	-1	-1	0	1	1	0

and the cycle values are periodic as well.

Thus Proposition 4.13 reduces the classification to congruence conditions.

The resulting classifications are:

$$\begin{aligned} C_n \text{ is pseudo-Gorenstein}^* &\iff n \equiv 1, 2, 5, 10 \pmod{12}, \\ P_n \text{ is pseudo-Gorenstein}^* &\iff n \equiv 0, 2, 9, 11 \pmod{12}. \end{aligned}$$

4.7 Graph Suspensions

4.7.1 C -suspensions

Definition 4.14 (C -suspension). Let

$$\emptyset \neq C \subseteq V(G).$$

The C -suspension

$$G(C)$$

is obtained from G by adjoining a new vertex z and joining z to every vertex of C .

Independent sets of $G(C)$ split according to whether they contain z .

This gives

$$P_{G(C)}(x) = P_G(x) + xP_{G-C}(x). \quad (4.4)$$

The point of this identity is that both $P_G(-1)$ and the multiplicity $M(G)$ can be tracked

under suspension.

4.7.2 Suspending over a vertex cover

Definition 4.15 (Vertex cover). A set

$$C \subseteq V(G)$$

is a *vertex cover* if every edge of G is incident to at least one vertex of C .

If C is a vertex cover and

$$S_0 = V(G) \setminus C, \quad s = |S_0|,$$

then S_0 is independent, so $G - C$ is edgeless.

Hence

$$P_{G-C}(x) = (1+x)^s,$$

and (4.4) becomes

$$P_{G(C)}(x) = P_G(x) + x(1+x)^s. \quad (4.5)$$

In particular, if

$$s \geq 1,$$

then

$$P_{G(C)}(-1) = P_G(-1).$$

Theorem 4.16 (Hibi–Kara–Vien). *Let C be a vertex cover of G such that*

$$0 < |V(G) \setminus C| < \alpha(G).$$

Then

$$G \text{ is pseudo-Gorenstein}^* \iff G(C) \text{ is pseudo-Gorenstein}^*.$$

More precisely, if

$$H = G(C), \quad s = |V(G) \setminus C|, \quad \alpha = \alpha(H) = \alpha(G),$$

then

$$P_H(-1) = P_G(-1)$$

and

$$h_{S[z]/I(H)}(t) = h_{S/I(G)}(t) + t(1-t)^{\alpha-s-1}. \quad (4.6)$$

Under the hypotheses of Theorem 4.16, this implies

$$h_\alpha(H) = h_\alpha(G).$$

4.7.3 Extremal values of s

The preservation theorem excludes the two extremal cases

$$s = 0 \quad \text{and} \quad s = \alpha(G).$$

Suppose first that

$$s = \alpha(G).$$

Then

$$\alpha(H) = \alpha(G) + 1$$

and

$$h_{S[z]/I(H)}(t) = (1 - t)h_{S/I(G)}(t) + t. \quad (4.7)$$

If G is pseudo-Gorenstein*, so that

$$h_{\alpha(G)}(G) = 1,$$

then the new leading coefficient is

$$h_{\alpha(H)}(H) = -1.$$

Thus this extremal suspension does not preserve the pseudo-Gorenstein* property.

For example, start with

$$G = C_5$$

and chooses a vertex cover whose complement has size

$$s = \alpha(C_5) = 2.$$

After adjoining the suspension vertex z , the graph H has an independent set of size 3, namely the two vertices outside the cover together with z , so

$$\alpha(H) = 3.$$

4.7.4 Full suspensions

A *full suspension* corresponds to

$$C = V(G),$$

hence

$$s = 0.$$

Full suspension does not preserve pseudo-Gorenstein* in general.

Writing

$$\widehat{G}$$

for the full suspension of G , the following classification is given for paths and cycles.

Theorem 4.17 (Hibi–Kara–Vien). *For full suspensions of cycles and paths,*

$$\begin{aligned} \widehat{C}_n \text{ is pseudo-Gorenstein}^* &\iff n \equiv 0 \pmod{12}, \\ \widehat{P}_n \text{ is pseudo-Gorenstein}^* &\iff n \equiv 1, 10 \pmod{12}. \end{aligned}$$

For comparison, the unsuspended families satisfy

$$\begin{aligned} C_n \text{ is pseudo-Gorenstein}^* &\iff n \equiv 1, 2, 5, 10 \pmod{12}, \\ P_n \text{ is pseudo-Gorenstein}^* &\iff n \equiv 0, 2, 9, 11 \pmod{12}. \end{aligned}$$

4.8 Questions and further directions

The degree formula

$$\deg h_{S/I(G)}(t) = \alpha(G) - M(G)$$

links the root -1 of $P_G(x)$ with the h -polynomial. Since $P_G(-1)$ is also the signed reduced Euler characteristic of the independence complex (up to the usual convention), the same evaluation appears in the combinatorial, topological, and algebraic descriptions.

Questions recorded at the end of the talk were:

- (i) understand $M(G)$ under other graph operations;
- (ii) find a graph-theoretic interpretation of

$$M(G) = 0, \quad \text{equivalently} \quad P_G(-1) \neq 0;$$

- (iii) classify graph families according to the behavior of $P_G(x)$ at $x = -1$.

Selected Attributions

Attributions recorded in these notes:

- Fröberg (1990): characterization of edge ideals with linear resolution via chordality of the complement.

- Hibi–Matsuda (2018): realizability of arbitrary positive pairs $(\text{reg}, \deg h)$ for monomial ideals.
- Hibi–Matsuda–Van Tuyl (2019): realizability for edge ideals and the inequality

$$\text{reg}(S/I(G)) + \deg h_{S/I(G)}(t) \leq n.$$

- Biermann–Kara–O’Keefe–Skelton–Sosa Castillo (2025): formulas for $\deg h$ for several graph families.
- Ene–Herzog–Hibi–Saeedi Madani: the pseudo-Gorenstein condition for standard graded Cohen–Macaulay algebras.
- Biermann–Chau–Kara–O’Keefe–Skelton–Sosa Castillo–Vien: the degree formula in Theorem 4.9.
- Hibi–Kara–Vien: the pseudo-Gorenstein* criterion and the suspension results described above.

5 The graded Möbius algebra of a matroid

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Abstract

The talk concerned the graded Möbius algebra associated to a matroid and its homological invariants. After recalling minimal graded free resolutions, Betti numbers, projective dimension, and Castelnuovo–Mumford regularity, the discussion related these algebraic invariants to the combinatorial structure of the underlying matroid.

5.1 Free resolutions and numerical invariants

Let

$$S = \mathbb{K}[x_1, \dots, x_n]$$

be a standard graded polynomial ring, let $I \subseteq S$ be a homogeneous ideal, and set

$$R = S/I.$$

Its minimal graded free resolution over S has the form

$$0 \longleftarrow R \longleftarrow F_0 \longleftarrow F_1 \longleftarrow \cdots \longleftarrow F_p \longleftarrow 0, \quad p = \text{pdim}_S R \leq n,$$

where

$$F_i \cong \bigoplus_j S(-i-j)^{\beta_{i,i+j}^S(R)}.$$

The integers

$$\beta_{i,i+j}^S(R)$$

are the graded Betti numbers, while

$$\beta_i^S(R) = \sum_j \beta_{i,i+j}^S(R)$$

are the total Betti numbers.

The usual Betti-table convention places $\beta_{i,i+j}$ in column i and row j . With this convention, the Castelnuovo–Mumford regularity is the largest nonzero row index:

$$\operatorname{reg}_S(R) = \max \{j : \beta_{i,i+j}^S(R) \neq 0 \text{ for some } i\},$$

whereas the projective dimension is the largest nonzero column index:

$$\operatorname{pdim}_S(R) = \max \{i : \beta_{i,i+j}^S(R) \neq 0 \text{ for some } j\}.$$

Example 5.1. Consider the Betti table

	0	1	2	3
0	1	.	.	.
1	.	2	.	.
2	.	1	1	.
3	.	.	2	.
4	.	.	.	1

Then

$$\operatorname{reg}_S(R) = 4, \quad \operatorname{pdim}_S(R) = 3.$$

If

$$S = \mathbb{K}[x, y, z],$$

the alternating sum of the free modules gives

$$H_R(t) = \frac{1 - (2t^2 + t^3) + (t^4 + 2t^5) - t^7}{(1-t)^3}.$$

Hence

$$H_R(t) = 1 + 3t + 4t^2 + \cdots.$$

Thus the same resolution simultaneously records homological invariants and the Hilbert series.

5.2 Koszul algebras

Let

$$R = \bigoplus_{i \geq 0} R_i$$

be a standard graded $\mathbb{K}$ -algebra and let

$$\mathfrak{m} = \bigoplus_{i \geq 1} R_i.$$

Definition 5.2. The algebra R is *Koszul* if the residue field

$$R/\mathfrak{m} \cong \mathbb{K}$$

has a linear minimal free resolution as an R -module.

Example 5.3. For

$$R = \mathbb{K}[x, y, z],$$

the residue field has the Koszul resolution

$$0 \longleftarrow \mathbb{K} \longleftarrow R \longleftarrow R(-1)^3 \longleftarrow R(-2)^3 \longleftarrow R(-3) \longleftarrow 0.$$

With the indicated bases, the differentials are

$$d_1 = \begin{pmatrix} x & y & z \end{pmatrix}, \quad d_2 = \begin{pmatrix} -y & -z & 0 \\ x & 0 & -z \\ 0 & x & y \end{pmatrix}, \quad d_3 = \begin{pmatrix} z \\ -y \\ x \end{pmatrix}.$$

Every differential is represented by linear forms. The Betti table therefore consists of a single row:

$$\begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 1 & 3 & 3 & 1 \end{array}.$$

5.2.1 When does quadratic imply Koszul?

A Koszul algebra must be quadratic, but the converse fails. Even among Gorenstein algebras presented by quadrics, the answer depends sharply on regularity and codimension.

Theorem 5.4 (Conca–Rossi–Valla, 2001). *Let R be a graded Gorenstein $\mathbb{K}$ -algebra presented by quadrics. If*

$$\operatorname{reg}(R) \leq 2,$$

or if

$$\operatorname{reg}(R) = 3 \quad \text{and} \quad \operatorname{codim}(R) \leq 4,$$

then R is Koszul.

Theorem 5.5 (Mastroeni–Schenck–Stillman, 2021/2023). *For every pair of integers*

$$c \geq r + 2 \geq 6,$$

there exists a graded quadratic Gorenstein $\mathbb{K}$ -algebra R satisfying

$$\operatorname{reg}(R) = r, \quad \operatorname{codim}(R) = c,$$

which is not Koszul.

There is also no universal finite test for Koszulness obtained by inspecting only a fixed initial segment of the resolution of the residue field.

Theorem 5.6 (Roos, 1993). *Koszulness of a standard graded algebra cannot, in general, be decided from finitely many prescribed steps of the minimal free resolution of its residue field.*

5.2.2 Restrictions imposed by Koszulness

The Poincaré series of R is

$$P_R(t) = \sum_{i \geq 0} \beta_i^R(\mathbb{K}) t^i,$$

the generating function for the total Betti numbers in the minimal R -free resolution of $\mathbb{K}$.

Proposition 5.7. *Let $R = S/I$ be a standard graded Koszul algebra. Then:*

(i) *I is generated by quadrics;*

(ii)

$$\operatorname{reg}_S(R) \leq \operatorname{pdim}_S(R);$$

(iii)

$$\beta_{i,i+j}^S(R) = 0 \quad \text{for } j > i;$$

(iv) if c is the number of quadratic minimal generators of I , then

$$\beta_{i,2i}^S(R) \leq \binom{c}{i};$$

(v) the Poincaré and Hilbert series satisfy

$$P_R(t)H_R(-t) = 1.$$

Two standard methods for proving Koszulness are particularly relevant here.

First, one may construct a *Koszul filtration*: a family of ideals generated by linear forms, containing 0 and $\mathfrak{m}$, and recursively stable under suitable colon operations. Such filtrations can be effective, although they are often large.

Second, one may prove that R is G -quadratic, meaning that its defining ideal admits a quadratic Gröbner basis. In that case the corresponding quadratic monomial initial algebra is Koszul, and the usual deformation argument yields the Koszulness of R .

5.3 Matroids and geometric lattices

5.3.1 A running example: the Fano matroid

The basic combinatorial example is the projective plane

$$PG(2, 2) = F_7.$$

It is represented over $\mathbb{F}_2$ by the columns of

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix},$$

which we label

$$p_1, \dots, p_7.$$

Its seven rank-two flats are

$$127, \quad 136, \quad 145, \quad 235, \quad 246, \quad 347, \quad 567,$$

where, for example,

$$127 = \{p_1, p_2, p_7\}.$$

Definition 5.8. A *matroid* is a pair

$$M = (E, \mathcal{I}),$$

where E is a finite ground set and

$$\mathcal{I} \subseteq 2^E$$

is the collection of independent sets, satisfying:

(i)

$$\emptyset \in \mathcal{I};$$

(ii) if

$$I' \subseteq I \quad \text{and} \quad I \in \mathcal{I},$$

then

$$I' \in \mathcal{I};$$

(iii) if

$$I, J \in \mathcal{I} \quad \text{and} \quad |I| < |J|,$$

then there exists

$$e \in J \setminus I$$

such that

$$I \cup \{e\} \in \mathcal{I}.$$

Linear independence in a vector space gives the prototypical example. Other standard sources include cycle matroids of graphs and matroids associated to hyperplane arrangements; Boolean matroids give the extremal free example.

Maximal independent sets are the *bases*. Minimal dependent sets are the *circuits*.

For $F \subseteq E$, the rank is

$$\text{rank}(F) = \max \{|I| : I \subseteq F, I \in \mathcal{I}\},$$

and the closure is

$$\text{cl}(F) = \{e \in E : \text{rank}(F \cup \{e\}) = \text{rank}(F)\}.$$

A set F is a *flat* if

$$\text{cl}(F) = F.$$

Example 5.9. For F_7 , one has

Set	Type and rank	Closure
$\{p_2, p_3, p_4\}$	basis, rank 3	$\{p_1, \dots, p_7\}$
$\{p_2, p_3\}$	independent, rank 2	$\{p_2, p_3, p_5\}$
$\{p_1, p_2, p_3, p_4\}$	circuit, rank 3	$\{p_1, \dots, p_7\}$
$\{p_3, p_4, p_7\}$	dependent flat, rank 2	$\{p_3, p_4, p_7\}$

5.3.2 Geometric lattices

Definition 5.10. A *lattice* is a poset in which any two elements F, G have a join

$$F \vee G$$

and a meet

$$F \wedge G.$$

A finite ranked lattice is called *geometric* if it is atomic and upper semimodular. In rank-function language, upper semimodularity is expressed by

$$\text{rank}(F \vee G) + \text{rank}(F \wedge G) \leq \text{rank}(F) + \text{rank}(G).$$

Theorem 5.11. *A finite lattice is geometric if and only if it is isomorphic to the lattice of flats of a matroid.*

For the Fano matroid, the lattice of flats has four ranks. Writing only the labels of the flats, its layers are

$$\begin{aligned} \text{rank } 3 & : 1234567, \\ \text{rank } 2 & : 127, 136, 145, 235, 246, 347, 567, \\ \text{rank } 1 & : 1, 2, 3, 4, 5, 6, 7, \\ \text{rank } 0 & : \emptyset. \end{aligned}$$

Thus the Whitney numbers of the second kind are

$$(1, 7, 7, 1).$$

5.3.3 Comparing the ranks

Let $\mathcal{L}$ be a geometric lattice of rank $n > 1$, and set

$$\mathcal{L}^k = \{F \in \mathcal{L} : \text{rank}(F) = k\}, \quad W_k = |\mathcal{L}^k|.$$

Theorem 5.12 (Greene, 1970). *For*

$$k = 2, \dots, n - 1,$$

one has

$$W_1 \leq W_k.$$

Moreover,

$$W_1 = W_k$$

if and only if

$$k = n - 1$$

and $\mathcal{L}$ is modular.

Here modularity means that for every

$$F, G \in \mathcal{L},$$

one has

$$\text{rank}(F \vee G) + \text{rank}(F \wedge G) = \text{rank}(F) + \text{rank}(G).$$

The stronger top-heavy statement compares all ranks symmetrically placed about the middle.

Theorem 5.13 (Huh–Wang, 2017; Braden–Huh–Matherne–Proudfoot–Wang, 2020).

Let $\mathcal{L}$ be a geometric lattice of rank r . Whenever

$$k \leq j \leq r - k,$$

there exists an injective map

$$\iota : \mathcal{L}^k \hookrightarrow \mathcal{L}^j.$$

In particular,

$$W_k \leq W_j \quad (k \leq j \leq r - k).$$

5.4 The graded Möbius algebra

Let M be a simple matroid with lattice of flats

$$\mathcal{L}(M).$$

Definition 5.14. The *graded Möbius algebra* of M is the standard graded $\mathbb{K}$ -algebra

$$B_M = \bigoplus_{F \in \mathcal{L}(M)} \mathbb{K}x_F,$$

where

$$\deg(x_F) = \text{rank}(F),$$

and multiplication is defined by

$$x_F x_G = \begin{cases} x_{F \vee G}, & \text{rank}(F \vee G) = \text{rank}(F) + \text{rank}(G), \\ 0, & \text{otherwise.} \end{cases}$$

The construction turns the rank data of the lattice directly into the Hilbert function.

Proposition 5.15. *Let M be a simple matroid.*

(i) *For each i ,*

$$\dim_{\mathbb{K}}(B_M)_i = |\{F \in \mathcal{L}(M) : \text{rank}(F) = i\}|.$$

Thus the Hilbert function of B_M is the rank-generating function of $\mathcal{L}(M)$.

(ii)

$$\text{reg}(B_M) = \text{rank}(M) = \text{rank } \mathcal{L}(M).$$

(iii) *If $E = E(M)$ and B_M is viewed as a quotient of the polynomial ring on one variable for each element of E , then*

$$\text{pdim}(B_M) = |E|.$$

For

$$M = F_7,$$

the Hilbert function is

$$1, 7, 7, 1.$$

Consequently,

$$\text{reg}(B_{F_7}) = 3, \quad \text{pdim}(B_{F_7}) = 7.$$

5.4.1 A presentation by circuits

For a subset

$$A \subseteq E(M),$$

write

$$y_A = \prod_{i \in A} y_i.$$

If

$$\mathcal{C}(M)$$

denotes the set of circuits of M , then a convenient presentation is the following.

Theorem 5.16 (LaClair–Mastroeni–McCullough–Peeva). *Let M be a simple matroid. Then*

$$B_M \cong \frac{\mathbb{K}[y_i : i \in E(M)]}{\left(y_i^2 : i \in E(M)\right) + \left(y_{C \setminus \{i\}} - y_{C \setminus \{j\}} : C \in \mathcal{C}(M), i, j \in C\right)}.$$

This presentation makes the connection with chordality transparent: a circuit of size d contributes relations of degree $d - 1$, and the possibility of reducing large circuits through smaller ones controls whether the defining ideal is quadratic or has a quadratic Gröbner basis.

5.4.2 The graphic case and the Orlik–Solomon parallel

When

$$M = M(G)$$

is the cycle matroid of a graph G , the graded Möbius algebra exhibits a close parallel with the Orlik–Solomon algebra A_M .

The relationships relevant here may be summarized schematically as

$$\begin{array}{ccc} A_M \text{ has a quadratic Gröbner basis} & \iff & M \text{ is supersolvable} \\ \updownarrow & & \updownarrow \\ A_M \text{ is Koszul} & & \\ \updownarrow & & \\ A_M \text{ is quadratic} & \iff & G \text{ is chordal,} \end{array}$$

while on the graded Möbius side one has

$$\begin{array}{ccc} B_M \text{ has a quadratic Gröbner basis} & \iff & M \text{ is strongly } T\text{-chordal} \\ \downarrow & & \updownarrow \\ B_M \text{ is Koszul} & \iff & G \text{ is strongly chordal} \\ \downarrow & & \\ B_M \text{ is quadratic} & \iff & G \text{ is chordal.} \end{array}$$

In particular, for graphic matroids, Koszulness of the graded Möbius algebra is characterized by strong chordality of the underlying graph.

5.5 Modularity, Gorensteinness, and projective geometries

A basic structural result identifies the Gorenstein graded Möbius algebras.

Theorem 5.17 (Greene, 1970; Maeno–Numata, 2016). *For a simple matroid M , the following are equivalent:*

- (i) *the lattice $\mathcal{L}(M)$ is modular;*
- (ii) *the rank-generating function of $\mathcal{L}(M)$ is symmetric;*
- (iii) *the Hilbert function of B_M is symmetric;*
- (iv) *B_M is Gorenstein.*

The inverse-system description is also useful. If

$$\mathcal{B}(M)$$

is the set of bases of M , then, with the polynomial variables acting by contraction on the dual polynomial ring, B_M may be realized as

$$B_M \cong \frac{\mathbb{K}[y_i : i \in E(M)]}{\text{Ann}\left(\sum_{B \in \mathcal{B}(M)} y_B\right)}.$$

The principal new result highlighted in the talk concerns finite projective geometries.

Theorem 5.18 (Grate–McCullough). *If M is a finite projective geometry, then*

$$B_M$$

is Koszul.

More concretely, for

$$M = PG(d, q),$$

the argument fits into the chain

$$PG(d, q) \implies M \text{ strongly } T\text{-chordal} \iff B_M \text{ is } G\text{-quadratic} \implies B_M \text{ is Koszul}.$$

The same projective geometries are chordal, which also yields the weaker conclusion that

$$B_M$$

is quadratic.

Since projective geometries have modular lattices of flats, their graded Möbius algebras are simultaneously Gorenstein and Koszul. Thus this family provides a natural class in which Gorensteinness and Koszulness coexist for combinatorial reasons.

5.6 Affine geometries

Finite affine geometries arise from projective geometries by deleting the points on a hyperplane at infinity.

The corresponding graded Möbius algebras behave differently from the projective case.

For

$$M = AG(d, q),$$

the results discussed in the talk give substantial evidence toward a general Koszulness statement.

In particular, for

$$q \geq 3,$$

the algebra B_M is quadratic. When q is odd, the affine geometry is chordal, and quadraticity also follows from the general chordality criterion.

The smallest instructive case is

$$AG(2, 3).$$

Its graded Möbius algebra is Koszul, although its natural presentation does not admit a quadratic Gröbner basis. Thus Koszulness and G -quadraticity already diverge within this family.

Conjecture 5.19 (Grate–McCullough). *Let*

$$M = AG(d, q), \quad d \geq 2, \quad q \geq 3.$$

Then the graded Möbius algebra

$$B_M$$

is Koszul.

The example $AG(2, 3)$ is useful for two reasons. It supports the general Koszulness conjecture while showing that a proof cannot simply proceed by establishing G -quadraticity in the natural circuit presentation.

5.7 Further directions

A remaining problem is to determine which matroids have Koszul graded Möbius algebras.

The families discussed in the talk can be summarized as follows:

Matroid family	Status of B_M
Cycle matroids $M(G)$	Koszul exactly when G is strongly chordal
Projective geometries $PG(n, q)$, $n \geq 2$	Koszul
Affine geometries $AG(n, q)$, $n \geq 2$, $q \geq 3$	Conjecturally Koszul; known to be quadratic
Sylvester matroids	Not Koszul in general
Lattice-path matroids	Open
Transversal matroids	Open
Sparse paving and paving matroids	Open
Dowling geometries	Open

These questions sit at the intersection of matroid structure and homological algebra. The graded Möbius algebra remembers the Whitney numbers through its Hilbert function, while its defining equations, Gröbner bases, and free resolution distinguish considerably finer combinatorial behavior.

The projective and affine families indicate that structural notions of matroid chordality should continue to play a central role in understanding Koszulness.

6 Residual Intersections and Their Connections to Linkage

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Abstract

The talk concerned residual intersections as a generalization of classical linkage. It reviewed the basic theory of residual intersections, compared their properties with those of linkage, and discussed several related questions and open problems.

6.1 Residual intersections and linkage

Residual intersection theory was introduced by Artin and Nagata as a natural generalization of classical linkage. Geometrically, if X is an algebraic variety and $Y \subseteq Z \subseteq X$ are closed subschemes, then the residual intersection of Y in Z is another closed subscheme W such that $W \cup Y = Z$. The algebraic theory studies the corresponding ideal-theoretic construction.

6.2 Basic definitions

Definition 6.1. Let R be a Noetherian ring and let $I \subseteq R$ be an ideal of height g . Let $\mathfrak{a} = (a_1, \dots, a_s) \subseteq I$, with $s \geq g$, and set

$$J = \mathfrak{a} : I.$$

Then J is called an *s-residual intersection* of I if $\text{ht}(J) \geq s$. It is called a *geometric s-residual intersection* if, in addition,

$$\text{ht}(I + J) > s.$$

The case $s = g$ is the one most closely related to classical linkage.

Example 6.2. Let I and J be prime ideals in a Noetherian ring R , with $\text{ht}(I) = g$, $I \not\subseteq J$, and suppose that $I \cap J$ is generated by s elements. If $\mathfrak{a} = I \cap J$, then $J = \mathfrak{a} : I$ is a geometric s -residual intersection of I .

Remark 6.3. Assume that R is local Gorenstein and that I is an unmixed ideal of height g . Then (geometric) linkage is the same as a (geometric) g -residual intersection.

6.3 Linkage and the Peskine–Szpiro theorem

A fundamental theorem of Peskine and Szpiro explains why linkage is so useful in the Cohen–Macaulay setting.

Theorem 6.4 (Peskine–Szpiro). *Let R be a local Gorenstein ring, and suppose that ideals I and J are directly linked by a complete intersection $\mathfrak{a} \subseteq I \cap J$; that is,*

$$J = \mathfrak{a} : I, \quad I = \mathfrak{a} : J.$$

Then the following hold.

- (a) *R/I is Cohen–Macaulay if and only if R/J is Cohen–Macaulay.*
- (b) *The canonical modules are given by*

$$\omega_{R/I} \cong J/\mathfrak{a}, \quad \omega_{R/J} \cong I/\mathfrak{a}.$$

This leads naturally to the following question.

Question 6.5. To what extent do the conclusions of the Peskine–Szpiro theorem remain valid for s -residual intersections when $s \geq g + 1$?

In general the answer is negative. Nevertheless, strong positive results are known in several important classes, including complete intersections, perfect ideals of grade 2, and Gorenstein ideals of grade 3.

6.4 The Artin–Nagata condition and residual S_2 properties

The basic homological conditions that arise in the subject are the following.

Definition 6.6. Let $(R, \mathfrak{m})$ be a local Cohen–Macaulay ring, let $I \subseteq R$ be an ideal of grade g , and let $s \geq g$.

- (i) We say that I satisfies the *Artin–Nagata condition* AN_s if for every integer i with $g \leq i \leq s$ and every i -residual intersection J of I , the quotient R/J is Cohen–Macaulay.
- (ii) We say that I is *s -residually S_2* if for every integer i with $g \leq i \leq s$ and every i -residual intersection J of I , the ring R/J satisfies Serre’s condition S_2 .
- (iii) We say that I is *weakly s -residually S_2* if the same condition holds for every *geometric* i -residual intersection J of I .

These conditions are designed precisely to control how much of the good behavior of linkage survives in higher-codimensional residual intersections.

6.5 Determinantal examples

The theory becomes especially concrete for determinantal ideals.

Theorem 6.7 (Eisenbud–Ulrich). *Let $\mathbb{k}$ be a field of characteristic 0, let*

$$R = \mathbb{k}[x_1, \dots, x_n, y_1, \dots, y_n], \quad X = \begin{pmatrix} x_1 & \cdots & x_n \\ y_1 & \cdots & y_n \end{pmatrix},$$

and let $I = I_2(X)$ be the ideal generated by the 2×2 minors of X . Let $\mathfrak{a} = (a_1, \dots, a_\ell)$ be a minimal reduction of I , where $\ell = \ell(I)$ is the analytic spread of I . For $1 \leq i \leq \ell$, set

$$\mathfrak{a}_i = (a_1, \dots, a_i), \quad K_i = \mathfrak{a}_i : I, \quad R_i = R/K_i.$$

If $g = \text{ht}(I) = n - 1$, then for every $i \geq g + 1$ one has

$$\text{depth } R_i = 2n - i - 3 = \dim R_i - 3.$$

Moreover, I is weakly $(\ell(I) - 2)$ -residually S_2 , and the canonical module of R_i is described by a power of the image of I in R_i :

$$\omega_{R_i} \cong (IR_i)^{i-g+1}(-2n),$$

up to the usual grading convention for twists.

A similar pattern persists for maximal minors of a generic $m \times n$ matrix: the canonical module of the corresponding residual intersection is controlled by an explicit power of the image of the determinantal ideal, together with the ambient grading shift.

6.6 Generators of residual intersections

A useful structural question is to determine how a residual intersection is generated.

Proposition 6.8. *Let R be a local Cohen–Macaulay ring and let $I = (x_1, \dots, x_g) \subseteq R$ be an ideal of grade g . Let $\mathfrak{a} = (a_1, \dots, a_s) \subseteq I$, and write*

$$(a_1, \dots, a_s) = [x_1 \cdots x_g]B$$

for a $g \times s$ matrix B with entries in R . If

$$J = \mathfrak{a} : I$$

is an s -residual intersection of I , then

$$I_g(B) \subseteq J,$$

and in fact

$$J = \mathfrak{a} + I_g(B).$$

In particular, when $g = 1$ the description is immediate, and for $g > 1$ the residual intersection is obtained by adjoining the maximal minors of the coefficient matrix B .

This description is one of the standard entry points for explicit calculations.

6.7 $(g + 1)$ -residual intersections as sums of links

The first genuinely new case beyond linkage occurs for $(g + 1)$ -residual intersections.

Definition 6.9. Let I be an ideal of grade g , let $\mathfrak{a} = (a_1, \dots, a_{g+1}) \subseteq I$, and set

$$J = \mathfrak{a} : I.$$

Assume that J is a $(g + 1)$ -residual intersection of I . We say that J is a *sum of links* if

$$J = \sum_{i=1}^{g+1} (a_1, \dots, \widehat{a_i}, \dots, a_{g+1}) : I.$$

Theorem 6.10 (Kustin–Ulrich). *Let R be a Gorenstein local ring and let $I \subseteq R$ be an ideal of grade g satisfying the standard residual-intersection hypotheses (for instance, in the classical situations of complete intersections, perfect ideals of grade 2, or Gorenstein ideals of grade 3). Then every $(g + 1)$ -residual intersection of I is a sum of links.*

Thus, even though a $(g + 1)$ -residual intersection need not itself be a direct link, it can still be decomposed into a controlled sum of ordinary links.

Example 6.11. Let

$$X = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \end{pmatrix}, \quad I = I_2(X),$$

and write

$$\Delta_{ij} = x_i y_j - x_j y_i.$$

Consider the ideal

$$\mathfrak{a} = (\Delta_{12}, \Delta_{13}, \Delta_{24}, \Delta_{34}) \subseteq I.$$

Then

$$J = \mathfrak{a} : I$$

is a 4-residual intersection of the height-3 ideal I , and it decomposes as a sum of links:

$$\begin{aligned} J = & (\Delta_{12}, \Delta_{13}, \Delta_{24}) : I + (\Delta_{12}, \Delta_{13}, \Delta_{34}) : I \\ & + (\Delta_{12}, \Delta_{24}, \Delta_{34}) : I + (\Delta_{13}, \Delta_{24}, \Delta_{34}) : I. \end{aligned}$$

Each summand is an ordinary link of I by one of the three-generated subideals of $\mathfrak{a}$, and their sum recovers the full residual intersection.

6.8 Remarks and open directions

Residual intersections extend linkage beyond the codimension-preserving complete-intersection case. The full Peskine–Szpiro package does not persist for arbitrary s -residual intersections, but analogues survive under the Artin–Nagata and residual S_2 hypotheses discussed above. Determinantal ideals supply the main test family in the talk, especially for canonical modules and explicit generators.

7 Quasi-Gorenstein Morphisms over DG-Rings

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Joint work with Andrew J. Soto Levins and Ryan Watson

Abstract

The talk extended quasi-Gorenstein homomorphisms from local rings to local DG-rings. After the necessary DG and derived-category background, the new ingredient was virtual G -smallness. The main result characterizes quasi-Gorenstein morphisms using bounded finite-length DG-modules of finite Gorenstein dimension.

7.1 Classical quasi-Gorenstein homomorphisms

Let

$$\varphi: (R, \mathfrak{m}, k) \longrightarrow (S, \mathfrak{n}, \ell)$$

be a finite local homomorphism of Noetherian local rings. Recall first the classical definition.

Definition 7.1 (Quasi-Gorenstein homomorphism). The map $\varphi: R \rightarrow S$ is *quasi-Gorenstein* if

$$\mathrm{Gdim}_R S < \infty$$

and there is an isomorphism, up to suspension, in the derived category of S ,

$$\mathbf{R}\mathrm{Hom}_R(S, R) \simeq \Sigma^a S$$

for some $a \in \mathbb{Z}$.

Two standard consequences were recalled.

Proposition 7.2. *Let $(R, \mathfrak{m}, k)$ be a Noetherian local ring. The canonical map*

$$R \longrightarrow k$$

is quasi-Gorenstein if and only if R is Gorenstein.

Proposition 7.3. *If $\varphi: R \rightarrow S$ is quasi-Gorenstein, then*

$$R \text{ is Gorenstein} \iff S \text{ is Gorenstein.}$$

Moreover, for a finitely generated S -module, and more generally for a bounded homo-

logically finite S -complex M , finiteness of Gorenstein dimension over R and over S agree:

$$\mathrm{Gdim}_R M < \infty \iff \mathrm{Gdim}_S M < \infty.$$

Thus quasi-Gorenstein maps behave as the natural morphisms along which Gorenstein finiteness is transported.

7.2 DG-rings and DG-modules

The extension of the preceding theory requires a derived replacement for a local ring.

Definition 7.4 (DG-algebra). A differential graded algebra is a graded algebra

$$R = \bigoplus_{i \in \mathbb{Z}} R_i$$

equipped with a differential

$$\partial_i: R_i \longrightarrow R_{i-1}$$

satisfying $\partial^2 = 0$ and the graded Leibniz rule

$$\partial(rs) = \partial(r)s + (-1)^{|r|}r\partial(s)$$

for homogeneous elements $r, s \in R$.

Throughout the talk the DG-algebras are strongly graded-commutative:

$$rs = (-1)^{|r||s|}sr,$$

and every homogeneous element of odd degree satisfies $r^2 = 0$.

The local DG-rings under consideration are connective and homologically finite in the following sense: $R_i = 0$ for $i < 0$, $H_0(R)$ is a Noetherian local ring, and $H(R)$ is finitely generated over $H_0(R)$. In particular, each $H_i(R)$ is a finitely generated $H_0(R)$ -module and $H_i(R) = 0$ for $i \gg 0$.

Definition 7.5 (DG-module). Let R be a DG-ring. A DG R -module is a graded R -module

$$M = \bigoplus_{i \in \mathbb{Z}} M_i$$

with a differential $\partial_M: M_i \rightarrow M_{i-1}$ satisfying

$$\partial_M(rm) = \partial_R(r)m + (-1)^{|r|}r\partial_M(m)$$

for homogeneous $r \in R$ and $m \in M$.

A morphism of local DG-rings

$$\varphi: R \longrightarrow S$$

is required to induce a local homomorphism

$$H_0(R) \longrightarrow H_0(S).$$

7.3 The derived category and derived functors

Write $D(R)$ for the derived category of DG R -modules. Quasi-isomorphic DG-modules become isomorphic in $D(R)$. The two basic derived functors are

$$- \otimes_R^{\mathbf{L}} - \quad \text{and} \quad \mathbf{R}\mathrm{Hom}_R(-, -).$$

They may be computed by replacing the relevant argument with a suitable semifree, projective, or injective resolution. Thus, if

$$P \xrightarrow{\sim} M, \quad F \xrightarrow{\sim} M, \quad N \xrightarrow{\sim} I$$

are appropriate resolutions, then the ordinary tensor and Hom constructions on these replacements compute the corresponding derived functors.

The cohomology of derived Hom recovers Ext:

$$H^i(\mathbf{R}\mathrm{Hom}_R(M, N)) \cong \mathrm{Ext}_R^i(M, N),$$

with the usual conversion between homological and cohomological grading conventions.

7.4 Projective dimension and Gorenstein dimension

One way to formulate projective dimension for a DG-module M is through eventual Ext-vanishing. In the bounded setting,

$$\mathrm{projdim}_R(M) = \inf \left\{ n \in \mathbb{Z} \mid \mathrm{Ext}_R^i(M, N) = 0 \text{ for all } N \in D_b(R) \text{ and all } i > n - \inf(N) \right\}.$$

Equivalently, finite projective dimension means that the functor

$$\mathbf{R}\mathrm{Hom}_R(M, -)$$

has finite homological displacement.

Gorenstein dimension is defined using derived biduality.

Definition 7.6 (Derived reflexivity). Let $M \in \mathbf{D}_b^f(R)$. The DG-module M is *derived reflexive* if

$$\mathbf{R}\mathrm{Hom}_R(M, R) \in \mathbf{D}_b^f(R)$$

and the canonical biduality morphism

$$M \longrightarrow \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(M, R), R)$$

is an isomorphism in $\mathbf{D}(R)$.

Definition 7.7 (Gorenstein dimension). For a derived-reflexive DG-module $M \in \mathbf{D}_b^f(R)$, set

$$\mathrm{Gdim}_R(M) = -\inf \mathbf{R}\mathrm{Hom}_R(M, R).$$

If M is not derived reflexive, then

$$\mathrm{Gdim}_R(M) = \infty.$$

This is the DG version of Auslander–Bridger Gorenstein dimension.

7.5 Quasi-Gorenstein morphisms of DG-rings

The classical definition extends directly once Gorenstein dimension and derived Hom have been set up in the DG category.

Definition 7.8 (Quasi-Gorenstein morphism). Let

$$\varphi: R \longrightarrow S$$

be a morphism of commutative local DG-rings. Assume

$$\mathrm{Gdim}_R S < \infty.$$

Then φ is *quasi-Gorenstein* if

$$\mathbf{R}\mathrm{Hom}_R(S, R) \simeq \Sigma^a S$$

in $\mathbf{D}(S)$ for some $a \in \mathbb{Z}$.

Thus the relative dualizing object $\mathbf{R}\mathrm{Hom}_R(S, R)$ is required to be nothing more than a shift of the target DG-ring itself. When R and S are ordinary Noetherian local rings and φ is module-finite, this recovers the classical definition.

7.6 Thick subcategories and virtual smallness

The main characterization is phrased in terms of thick subcategories of the derived category.

Definition 7.9 (Thick subcategory). A full subcategory $\mathcal{C} \subseteq \mathbf{D}(R)$ is *thick* if it is closed under isomorphisms, shifts, finite direct sums, direct summands, and distinguished triangles.

For $M \in \mathbf{D}(R)$, write

$$\mathrm{Thick}_R(M)$$

for the smallest thick subcategory of $\mathbf{D}(R)$ containing M .

Two basic examples are important:

$$\mathrm{Thick}_R(R) = \{\text{perfect DG } R\text{-modules}\},$$

and, in the local Noetherian setting,

$$\mathrm{Thick}_R(k) = \{M \in \mathbf{D}_b^f(R) : H(M) \text{ has finite length}\}.$$

Definition 7.10 (Virtually small). An object $M \in \mathbf{D}(R)$ is *virtually small* if

$$\mathrm{Thick}_R(M) \cap \mathrm{Thick}_R(R) \neq \{0\}.$$

Equivalently, M finitely builds a nonzero perfect object.

For the Gorenstein analogue, replace perfect objects by objects of finite Gorenstein dimension.

Definition 7.11 (Virtually G -small). Let $M \in \mathbf{D}_b^f(R)$. The DG-module M is *virtually G -small* if there exists a nonzero object

$$F_M \in \mathrm{Thick}_R(M)$$

with

$$\mathrm{Gdim}_R(F_M) < \infty.$$

Equivalently, the thick subcategory generated by M meets the class of nonzero derived-reflexive objects.

This definition is designed so that finite Gorenstein dimension plays the role that finite projective dimension plays in ordinary virtual smallness.

7.7 The complete-intersection analogy

The motivation for the main theorem comes from the corresponding detection theorem for complete-intersection maps. A result of Briggs, Iyengar, Letz, and Pollitz gives a model for what one should expect.

Theorem 7.12 (Briggs–Iyengar–Letz–Pollitz). *Let $\varphi: R \rightarrow S$ be a surjective local homomorphism of Noetherian local rings of finite flat dimension. Then φ is a complete-intersection homomorphism if and only if every S -complex with bounded finite-length homology that is proxy small over R is proxy small over S .*

The guiding analogy is therefore

finite projective dimension	$\longleftrightarrow$	finite Gorenstein dimension,
complete-intersection map	$\longleftrightarrow$	quasi-Gorenstein map,
proxy/virtual smallness	$\longleftrightarrow$	virtual G -smallness.

7.8 The main characterization

Let

$$\mathbf{D}_b^{\text{fl}}(S) = \left\{ M \in \mathbf{D}_b^f(S) \mid H(M) \text{ has finite length over } H_0(S) \right\}.$$

The main theorem gives the following characterization in terms of virtual G -smallness.

Theorem 7.13 (Nason–Soto Levins–Watson, 2026). *Let*

$$\varphi: R \longrightarrow S$$

be a morphism of commutative local DG-rings such that S_0 is a Noetherian local ring and

$$\text{Gdim}_R S < \infty.$$

Then the following are equivalent:

- (i) *φ is quasi-Gorenstein;*
- (ii) *every DG S -module*

$$M \in \mathbf{D}_b^{\text{fl}}(S)$$

with

$$\text{Gdim}_R M < \infty$$

is virtually G -small over S .

The forward implication reflects the preservation of finite Gorenstein dimension along a

quasi-Gorenstein morphism: an S -object having finite Gorenstein dimension over R also has finite Gorenstein dimension over S , so it already supplies the required finite-Gorenstein-dimension object in its own thick closure. The converse is the substantive direction: the finite-length test objects force the relative dualizing complex

$$\mathbf{R}\mathrm{Hom}_R(S, R)$$

to have the rigidity required to be a shift of S .

The theorem is new even when both R and S are ordinary rings. It gives a Gorenstein counterpart to the complete-intersection detection theorem above and shows that virtual G -smallness is the correct categorical finiteness condition for quasi-Gorenstein morphisms.

Recap

For a local DG-ring morphism $\varphi: R \rightarrow S$ of finite relative Gorenstein dimension, quasi-Gorenstein means

$$\mathbf{R}\mathrm{Hom}_R(S, R) \simeq \Sigma^a S$$

for some a . The characterization in the talk tests this condition on bounded DG S -modules with finite-length homology: finite Gorenstein dimension over R forces virtual G -smallness over S . This is the Gorenstein analogue of proxy-small detection for complete-intersection morphisms.

8 Characterizations of Minimal Frobenius Splitting Numbers in Gorenstein Rings

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Abstract

The talk studied the minimal positive Frobenius splitting numbers of an F -finite, F -pure Gorenstein local ring. The equivalent conditions involve the splitting ideals $I_e(R)$ and the splitting prime $\mathcal{P}(R)$. In the Gorenstein case the splitting ideals can be written by colon ideals using a parameter ideal and a socle generator; this gives the hypersurface and perturbation criteria. A Veronese example shows where the propagation statement fails without Gorensteinness.

8.1 Frobenius and restriction of scalars

Throughout this lecture, $(R, \mathfrak{m}, k)$ is a commutative Noetherian local ring of characteristic $p > 0$, and

$$d = \dim R.$$

For each $e \geq 1$, write $q = p^e$. The e th iterate of the Frobenius endomorphism is

$$F^e: R \longrightarrow R, \quad r \longmapsto r^q.$$

The module $F_*^e R$ is obtained from R by restriction of scalars along F^e . Thus $F_*^e R$ agrees with R as an abelian group, but its R -module structure is given by

$$r \cdot F_*^e s = F_*^e(r^q s) \quad (r, s \in R).$$

Definition 8.1 (F -finite ring). The ring R is F -finite if $F_* R$ is a finitely generated R -module. Equivalently, $F_*^e R$ is finitely generated for every $e \geq 1$.

Example 8.2 (Polynomial rings over a perfect field). Let

$$R = k[x_1, \dots, x_d]$$

with k perfect. Then $F_*^e R$ is free of rank $q^d = p^{ed}$, with R -basis

$$\left\{ F_*^e(x_1^{i_1} \cdots x_d^{i_d}) \mid 0 \leq i_j < q \right\}.$$

Thus, in the regular case, every summand of $F_*^e R$ is free.

The preceding example is a special case of Kunz's theorem.

Theorem 8.3 (Kunz). *Let R be a Noetherian local ring of characteristic $p > 0$. Then R is regular if and only if Frobenius is flat. In the F -finite local setting, this is equivalent to $F_*^e R$ being a free R -module for some, equivalently every, $e \geq 1$.*

8.2 F -purity, strong F -regularity, and splitting numbers

Two Frobenius singularity classes used in the talk are:

Definition 8.4 (F -purity). An F -finite local ring R is F -pure if the Frobenius map

$$R \longrightarrow F_* R, \quad 1 \longmapsto F_* 1,$$

splits as a map of R -modules. Equivalently, there is an R -linear map

$$\phi: F_* R \longrightarrow R$$

with $\phi(F_* 1) = 1$.

Definition 8.5 (Strong F -regularity). Put

$$R^\circ = R \setminus \bigcup_{\mathfrak{p} \in \text{Min } R} \mathfrak{p}.$$

The ring R is *strongly F -regular* if, for every $c \in R^\circ$, there is an integer $e \geq 1$ such that the map

$$R \longrightarrow F_*^e R, \quad 1 \longmapsto F_*^e c,$$

splits. Equivalently, there is some

$$\phi \in \text{Hom}_R(F_*^e R, R)$$

for which $\phi(F_*^e c) = 1$.

Even when R is singular, $F_*^e R$ may contain a substantial free direct summand. This leads to the numerical invariants at the center of the lecture.

Definition 8.6 (Frobenius splitting numbers). For every $e \geq 1$, write

$$F_*^e R \cong R^{\oplus a_e(R)} \oplus M_e,$$

where M_e has no free direct summand and $a_e(R)$ is maximal. The integer $a_e(R)$ is the

eth Frobenius splitting number of R .

When the residue field is perfect, the F -signature is

$$s(R) = \lim_{e \rightarrow \infty} \frac{a_e(R)}{p^{ed}}.$$

The limit exists for F -finite local rings. The fundamental extremal properties are

$$s(R) > 0 \iff R \text{ is strongly } F\text{-regular,}$$

and

$$s(R) = 1 \iff R \text{ is regular.}$$

Thus the free rank of Frobenius detects both regularity and strong F -regularity asymptotically.

There is also an asymptotic invariant adapted to F -pure rings that are not strongly F -regular. If $\mathcal{P}(R)$ denotes the splitting prime defined below and k is perfect, set

$$n = \dim(R/\mathcal{P}(R)).$$

This integer is the *Frobenius splitting dimension*. The corresponding splitting ratio is

$$r_F(R) = \lim_{e \rightarrow \infty} \frac{a_e(R)}{p^{en}},$$

and for an F -pure ring one has

$$0 < r_F(R) \leq 1.$$

When $n = d$, the splitting prime is zero and the splitting ratio agrees with the F -signature.

8.3 Splitting ideals and the splitting prime

The splitting numbers can be recovered from a family of ideals measuring which elements fail to produce a Frobenius splitting.

Definition 8.7 (Frobenius splitting ideal). For $e \geq 1$, define

$$I_e(R) = \{r \in R \mid \phi(F_*^e r) \in \mathfrak{m} \text{ for every } \phi \in \operatorname{Hom}_R(F_*^e R, R)\}.$$

Thus $r \notin I_e(R)$ precisely when some R -linear map

$$\phi: F_*^e R \longrightarrow R$$

sends $F_*^e r$ to a unit. After rescaling ϕ , this is equivalent to the map

$$R \longrightarrow F_*^e R, \quad 1 \longmapsto F_*^e r,$$

splitting.

A standard length formula relates $I_e(R)$ to the free rank:

$$a_e(R) = [k^{1/q} : k] \ell_R(R/I_e(R)), \quad q = p^e. \quad (8.1)$$

In particular, if R is F -pure, then the smallest possible value of $a_e(R)$ is

$$[k^{1/q} : k],$$

and this minimum occurs exactly when

$$I_e(R) = \mathfrak{m}.$$

When k is perfect, the minimum is simply 1.

Definition 8.8 (Splitting prime). Assume that R is F -pure. The *splitting prime* is

$$\mathcal{P}(R) = \bigcap_{e \geq 1} I_e(R).$$

It is a prime ideal. Moreover,

$$R \text{ is strongly } F\text{-regular} \iff \mathcal{P}(R) = 0.$$

At the opposite extreme, the condition

$$\mathcal{P}(R) = \mathfrak{m}$$

means that the Frobenius splitting dimension is zero. The main theorem identifies this condition with minimal Frobenius splitting numbers in the Gorenstein setting.

8.4 The Gorenstein colon formula

Suppose now that $(R, \mathfrak{m}, k)$ is a d -dimensional Gorenstein local ring. Let

$$\underline{x} = x_1, \dots, x_d$$

be a system of parameters and write

$$(\underline{x}) = (x_1, \dots, x_d).$$

Since R is Cohen–Macaulay, $R/(\underline{x})$ is Artinian Gorenstein, so its socle is one-dimensional over k . Choose $u \in R$ whose image generates

$$\text{soc}(R/(\underline{x})).$$

Equivalently,

$$(\underline{x}) :_R \mathfrak{m} = (\underline{x}, u). \quad (8.2)$$

For an ideal $J \subseteq R$ and $q = p^e$, write

$$J^{[q]} = (r^q : r \in J).$$

The Gorenstein hypothesis gives an explicit description of every splitting ideal.

Proposition 8.9 (Splitting ideals in the Gorenstein case). *With the notation above, for every $e \geq 1$ and $q = p^e$,*

$$I_e(R) = ((\underline{x})^{[q]} :_R u^q). \quad (8.3)$$

This formula is the computational input behind the hypersurface statements later in the lecture. It translates a condition involving all Cartier maps

$$F_*^e R \longrightarrow R$$

into a single colon ideal.

8.5 Hypersurface quotients

The familiar Fedder criterion is the basic model for the hypersurface calculations.

Criterion 8.10 (Fedder, hypersurface form). *Let $(S, \mathfrak{n}, k)$ be an F -finite regular local ring of characteristic $p > 0$, and let $0 \neq f \in \mathfrak{n}$. Then*

$$S/(f) \text{ is } F\text{-pure} \iff f^{p-1} \notin \mathfrak{n}^{[p]}.$$

For a Gorenstein ambient ring, the corresponding statement is encoded by the splitting ideals. Let $f \in R$ be a nonzerodivisor. A Frobenius map on $R/(f)$ is obtained from a map on R by multiplication by

$$f^{q-1}.$$

More precisely, the map determining splitting on the quotient is represented upstairs by

$$F_*^e R \xrightarrow{F_*^e(f^{q-1}, -)} F_*^e R \xrightarrow{\phi} R.$$

Consequently,

$$I_e(R/(f)) = \frac{(I_e(R) :_R f^{q-1})}{(f)}. \quad (8.4)$$

Combining (8.4) with the Gorenstein colon formula (8.3) gives

$$I_e(R/(f)) = \frac{(((\underline{x})^{[q]} :_R u^q) :_R f^{q-1})}{(f)}. \quad (8.5)$$

This colon description is the point where Gorensteinness enters the quotient calculation.

8.6 Minimal Frobenius splitting numbers

The main theorem gives the following equivalent conditions for minimal positive splitting numbers.

Theorem 8.11 (Akter: characterization of minimal splitting numbers). *Let $(R, \mathfrak{m}, k)$ be an F -finite, F -pure, Gorenstein local ring of characteristic $p > 0$. Then the following are equivalent.*

(1)

$$a_1(R) = [k^{1/p} : k].$$

(2) *There exists $e \geq 1$ such that*

$$a_e(R) = [k^{1/p^e} : k].$$

(3) *For every $e \geq 1$,*

$$a_e(R) = [k^{1/p^e} : k].$$

(4)

$$I_1(R) = \mathfrak{m}.$$

(5) *There exists $e \geq 1$ such that*

$$I_e(R) = \mathfrak{m}.$$

(6) *For every $e \geq 1$,*

$$I_e(R) = \mathfrak{m}.$$

(7)

$$\mathcal{P}(R) = \mathfrak{m}.$$

If k is perfect, conditions (1)–(3) simply say, respectively,

$$a_1(R) = 1, \quad a_e(R) = 1 \text{ for some } e, \quad a_e(R) = 1 \text{ for every } e.$$

Structure of the proof. The equivalence between the splitting-number conditions and the corresponding splitting-ideal conditions follows immediately from (8.1): the positive minimum of $a_e(R)$ is attained exactly when

$$\ell_R(R/I_e(R)) = 1,$$

which is equivalent to $I_e(R) = \mathfrak{m}$.

The genuinely Gorenstein step is the propagation of the condition $I_e(R) = \mathfrak{m}$ from one Frobenius level to all levels. In the Gorenstein case, the Cartier module

$$\mathrm{Hom}_R(F_*^e R, R)$$

is cyclic over $F_*^e R$, and its generators may be chosen compatibly under Frobenius iteration. This compatibility forces maximality of one splitting ideal to propagate to $I_1(R)$ and then to every $I_e(R)$. Finally,

$$\mathcal{P}(R) = \bigcap_{e \geq 1} I_e(R),$$

so condition (6) implies (7), while (7) forces every $I_e(R)$ to equal $\mathfrak{m}$ because each $I_e(R)$ is contained in $\mathfrak{m}$. $\square$

There is a useful elementary inequality among the splitting numbers that explains one direction of the theorem without using Gorenstein duality.

Lemma 8.12 (Supermultiplicativity of splitting numbers). *For all $e, f \geq 1$,*

$$a_{e+f}(R) \geq a_e(R)a_f(R).$$

Proof. Write

$$F_*^e R \cong R^{\oplus a_e(R)} \oplus M_e.$$

Applying F_*^f gives

$$F_*^{e+f} R \cong (F_*^f R)^{\oplus a_e(R)} \oplus F_*^f M_e.$$

Since each copy of $F_*^f R$ contains $R^{\oplus a_f(R)}$ as a direct summand, the first term contributes at least

$$a_e(R)a_f(R)$$

free summands to $F_*^{e+f} R$. $\square$

Corollary 8.13. *Assume that R is F -pure and that k is perfect. If*

$$a_e(R) = 1$$

for some $e \geq 1$, then

$$a_1(R) = 1.$$

Proof. By Lemma 8.12,

$$a_e(R) \geq a_1(R)^e.$$

Since F -purity gives $a_1(R) \geq 1$, the equality $a_e(R) = 1$ forces $a_1(R) = 1$. □

The converse implication

$$a_1(R) = 1 \implies a_e(R) = 1 \text{ for all } e$$

is therefore the subtle direction, and it is precisely where the Gorenstein hypothesis is used.

8.7 A hypersurface criterion

The main theorem combines with the colon formula (8.5) to give an explicit criterion for hypersurface quotients.

Corollary 8.14 (Minimal splitting numbers for a hypersurface). *Let $(R, \mathfrak{m}, k)$ be an F -finite, F -pure, Gorenstein local ring of dimension d and characteristic $p > 0$. Let $f \in \mathfrak{m}$ be a nonzerodivisor. Choose a system of parameters*

$$\underline{x} = x_1, \dots, x_d$$

and an element $u \in R$ satisfying

$$(\underline{x}) :_R \mathfrak{m} = (\underline{x}, u).$$

Then the following are equivalent.

(1)

$$a_1(R/(f)) = [k^{1/p} : k].$$

(2) *There exists $e \geq 1$ such that*

$$a_e(R/(f)) = [k^{1/p^e} : k].$$

(3) For every $e \geq 1$,

$$a_e(R/(f)) = [k^{1/p^e} : k].$$

(4)

$$\left(((\underline{x})^{[p]} :_R u^p) :_R f^{p-1} \right) = \mathfrak{m}.$$

(5) There exists $e \geq 1$ such that

$$\left(((\underline{x})^{[p^e]} :_R u^{p^e}) :_R f^{p^e-1} \right) = \mathfrak{m}.$$

(6) For every $e \geq 1$,

$$\left(((\underline{x})^{[p^e]} :_R u^{p^e}) :_R f^{p^e-1} \right) = \mathfrak{m}.$$

(7) The splitting prime of $R/(f)$ is the maximal ideal:

$$\mathcal{P}(R/(f)) = \mathfrak{m}/(f).$$

Moreover, if these equivalent conditions hold and

$$\epsilon \in (\underline{x})^{[p]},$$

then

$$\mathcal{P}(R/(f + \epsilon)) = \mathfrak{m}/(f + \epsilon).$$

In particular, when the quotient has splitting dimension zero, perturbations by the indicated Frobenius power of the parameter ideal preserve the condition.

8.8 Perturbations and continuity

The hypersurface result fits into a broader perturbation problem: how much of a singularity is unchanged after replacing a defining equation f by $f + \epsilon$ with ϵ very deep in the maximal ideal?

A classical theorem of Samuel provides a strong answer for isolated hypersurface singularities. Let

$$S = k[[x_1, \dots, x_d]], \quad \mathfrak{n} = (x_1, \dots, x_d),$$

and let

$$J(f) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_d} \right)$$

be the Jacobian ideal. If $J(f)$ is $\mathfrak{n}$ -primary and

$$\epsilon \in \mathfrak{n}J(f)^2,$$

then there is an automorphism of S carrying f to $f + \epsilon$. Consequently,

$$S/(f) \cong S/(f + \epsilon).$$

Exact isomorphism is much stronger than what is generally available, so one may instead ask whether numerical invariants vary continuously. The F -signature has this property at reduced parameter elements: in the Cohen–Macaulay F -finite setting, Polstra and Smirnov proved that for every $\delta > 0$, sufficiently small $\mathfrak{m}$ -adic perturbations satisfy

$$\left| s(R/(f)) - s(R/(f + \epsilon)) \right| < \delta.$$

The results above sharpen this perspective in the minimal-splitting situation: under the Gorenstein hypotheses, one obtains exact persistence of the splitting-prime condition for the explicit class of perturbations

$$\epsilon \in (\underline{x})^{[p]}.$$

8.9 Why the Gorenstein hypothesis matters

The propagation in Theorem 8.11 fails without the Gorenstein hypothesis; the example in the talk was a Veronese ring.

Example 8.15 (A non-Gorenstein Veronese ring). Let k be a perfect field of characteristic 2, let

$$S = k[x, y],$$

and let

$$R = S^{(3)} = \bigoplus_{n \geq 0} S_{3n}$$

be the third Veronese subring, localized at its homogeneous maximal ideal when viewed in the local setting. As an R -module, the polynomial ring decomposes according to total degree modulo 3:

$$S = R \oplus M_1 \oplus M_2,$$

where

$$\begin{aligned} M_1 &= \bigoplus_{a+b \equiv 1 \pmod{3}} kx^a y^b, \\ M_2 &= \bigoplus_{a+b \equiv 2 \pmod{3}} kx^a y^b. \end{aligned}$$

Thus R is a direct summand of the regular ring S , and in this tame situation it is strongly F -regular.

The Goto–Watanabe criterion for Veronese subrings says that the n th Veronese of a

polynomial ring in d variables is Gorenstein precisely when n divides d . Here

$$n = 3, \quad d = 2,$$

so R is not Gorenstein.

The splitting-number computation recorded in the lecture may be expressed as

$$a_e(R) = \# \left\{ (a, b) \in \{0, \dots, 2^e - 1\}^2 \mid a + b \equiv 0 \pmod{3} \right\}.$$

For $e = 1$, the only admissible pair is $(0, 0)$, so

$$a_1(R) = 1.$$

For $e = 2$, the admissible pairs have sums 0, 3, or 6. There are respectively 1, 4, and 1 such pairs, giving

$$a_2(R) = 6.$$

Hence

$$a_1(R) = 1 \quad \text{but} \quad a_2(R) > 1.$$

Thus the implication from a minimal first splitting number to minimal splitting numbers at every Frobenius level is false without the Gorenstein hypothesis.

Remark 8.16. The inequality in Lemma 8.12 is perfectly compatible with this example: it only gives

$$a_2(R) \geq a_1(R)^2 = 1.$$

The Gorenstein structure supplies the additional rigidity needed to force equality at all levels.

Recap

For an F -finite, F -pure Gorenstein local ring, the minimal positive value of $a_e(R)$ is $[k^{1/p^e} : k]$ (equal to 1 for a perfect residue field), and

$$a_e(R) = [k^{1/p^e} : k] \ell_R(R/I_e(R)).$$

Hence minimality is equivalent to $I_e(R) = \mathfrak{m}$. With a parameter ideal $(\underline{x})$ and a socle generator u modulo $(\underline{x})$, the Gorenstein formula

$$I_e(R) = ((\underline{x})^{[p^e]} :_R u^{p^e})$$

identifies this condition with $\mathcal{P}(R) = \mathfrak{m}$ and gives the propagation across Frobenius iterates. For $R/(f)$, the corresponding formula is

$$I_e(R/(f)) = \frac{(((\underline{x})^{[p^e]} :_R u^{p^e}) :_R f^{p^e-1})}{(f)}.$$

This is the input for the perturbation theorem. The third Veronese example in characteristic 2 has $a_1(R) = 1$ but $a_2(R) = 6$, showing that the Gorenstein hypothesis is essential.